

Alternative Manure Management + Bovaer for Dairy Protocol

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1 Introduction

This protocol credits the absolute GHG reductions that result from the combination dairy manure management practices that reduce the amount of CH₄ (and potentially N₂O, depending on the practice) emitted from manure collection, treatment, and/or storage and the use of Bovaer in lactating dairy cattle to reduce the amount of CH₄ from enteric emission.

When volatile manure solids are stored in anaerobic conditions, a significant amount of methane (CH₄) and nitrous oxide (N₂O) are produced. Implementing practices that redirect the treatment of volatile manure solids from anaerobic conditions to aerobic conditions, reduces the amount of volatile manure solids managed anaerobically and results in decreases in CH₄ and N₂O emissions. Manure that is handled as a solid or is deposited on land decomposes under more aerobic conditions, causing significantly less methane production plus a small amount of nitrous oxide. Manure that is collected and/or separated can be a component of a project, but this practice must be combined with a listed treatment and/or storage practice to be eligible, since methane emissions predominantly occur during the storage and/or treatment phase.

When anaerobic storage conditions exist at a dairy facility and they implement one of the practices included in this protocol, the facility can generate impact units that can be sold to organizations looking to reduce their GHG emissions. These organizations include any offtake partners such as processors of their milk for milk products, consumer packaged goods producers (CPG's) and other retailers, all three of whom might purchase the credits and otherwise encourage the producer to implement the practices in question.

Bovaer Quantification Methodology was developed by Elanco to provide a standardized methodology for the quantification of greenhouse gas (GHG) reductions associated with the use of 3-nitrooxypropanol (3-NOP) as a feed ingredient for U.S. lactating dairy cattle. The quantification of these reduction credits enables consumer packaged goods companies (CPG) to reduce their scope 3 emissions.

2 Project Definition

Bovaer is a feed ingredient for lactating dairy cattle that reduces enteric methane. Bovaer must be fed according to label instructions. The Bovaer label dose was determined by the Food and Drug Administration in May of 2024 to be 60-80 mg/kg based on both effectiveness and safety data. Any FDA-reviewed and accepted changes made to the label dose will be immediately communicated to key stakeholders (including nutritionists, producers, Athian, verifiers, and UpLook personnel) and updated in UpLook, including updates to QAQC procedures in place to ensure adherence to label (i.e., IF/THEN statements where if the operation reports dose levels outside the label range, the calculated emissions reductions equal zero.) Any changes made to the label dose must be legally followed by the farm. Grazing lactating dairy cattle are ineligible for this program. The only activity associated with this intervention is the feeding of Bovaer daily to lactating dairy cows and all other business practices remain unchanged.

Each project must also include at least one of the following eligible practices that reduce GHG emissions from manure management. Definitions of each eligible practice can be found in the glossary.

1. Pasture-based management including:
 - a. conversion of a non-pasture dairy operation to pasture-based management; or
 - b. at an existing pasture operation, increasing the amount of time livestock spend at pasture.

Note: Pasture-based management projects must have previously managed or stored some manure in anaerobic conditions and introduce practices that reduce the quantity of manure managed anaerobically.

2. Alternative manure treatment and storage including:
 - a. Installation of a compost bedded pack barn to compost manure on-site;
 - b. Installation of slatted floor pit storage manure collection, which must be cleaned out at least monthly.

Note: If pit storage is cleaned out less than twelve times per year, it does not qualify.

3. Conversion from a flush to scrape manure collection system in combination with one of the following practices:
 - a. Open solar drying of manure: manure is dried in a paved or unpaved open confinement area without any significant vegetative cover where accumulating manure may be removed periodically.
 - b. Closed solar drying: drying of manure in an enclosed environment
 - c. Forced evaporation with natural gas fueled dryers
 - d. Daily spread: manure is routinely removed from a confinement facility and is applied to cropland or pasture within 24 hours of excretion.
 - e. Solid storage: manure is stored in unconfined piles or stacks
 - f. Composting in vessel, either:
 - i. in an enclosed vessel, with forced aeration and continuous mixing; or
 - ii. in an aerated static pile: composting in piles with forced aeration but no mixing.
 - g. Composting in windrows, either:
 - i. Intensive windrows: turned at least daily for aeration;
 - ii. Passive windrows: infrequent turning
 - h. Solid separation with an aerated vermifiltration system.

Note: vermicomposting systems must be used in tandem with a solid separation device.

4. Solid separation of manure solids before being deposited into an anaerobic environment, paired with one of the practices (a) through (g) in the list above. Solid separation technologies include weeping walls, stationary screens, vibrating screens, screw presses, centrifuges, roller drums, belt presses/screens, advanced solid-liquid separation assisted by flocculants and/or bead filters, and vermifiltration. Additional technologies may be applicable, upon review.

Note: Either the installation of a new solid separation system that does not currently employ solid separation, or the installation of a new solid separation system with higher separation efficiency than the existing solid separation technology may be eligible.

2.1 Impact on Yield

There is no impact of Bovaer on dry matter intake, digestibility, or energy corrected milk. This is evidenced by several meta-analyses which are summarized in **Table 2.1**

Table 2.1 Impact of Bovaer on methane, dry matter intake, digestibility, and performance

	Methane	Dry Matter Intake	Milk Yield	Adjusted Milk Yield ¹	Milk Fat	Milk Protein	Digestibility ²
Jayanegara et al, 2018	-19%	=	=	=	Yield: NR %: ↑	Yield: NR %: =	=
Dijkstra et al., 2018	-39%	NR	NR	NR	NR	NR	NR
Kim et al., 2020	Linear ↓	=	=	=	Yield: NR %: =	Yield: NR %: =	=
Hristov et al., 2022	-28%	=	=	=	Yield: NR %: ↑	Yield: = %: =	NR
Kebreab et al., 2023	-32%	NR	NR	NR	NR	NR	NR
NR = not reported							
¹ Adjusted for components as either energy-corrected milk, fat-corrected milk, or solids-corrected milk							
² Includes dry matter, organic matter, and neutral detergent fiber (NDF) digestibility							

There is also no anticipated effect on yield or productivity associated with the manure management aspects of this program. Any changes in yield will be a result of changes in dry matter intake (DMI), which are measured in both the baseline and project scenarios. Proponents are discouraged from increasing milk yield because an increase in DMI would increase the emissions in the project scenario and, therefore, decrease the GHG reductions of the project.

2.2 Causality

Causality lies in that the funding from the sale of impact units will help not only recoup the initial capital cost of implementing eligible practices, but long term to sustain the operations and maintenance and improvement of those practices. The sale of impact units on a cadence that more closely matches the flow of business and cash flow needs of the producer allows for reduced risk in the upfront capital necessary to implement new practices and to ensure their reductions to GHG outputs of the farm continue into the future.

The sale of impact units also aims to encourage the long-term maintenance of alternative manure management practices implemented under this program. This maintenance includes equipment repair and maintenance costs, energy and fuel costs, labor, and business management costs. These costs are rarely, if ever, reflected in a premium price for the product. Without continued incentives or compensation, many of these practices are not financially viable long term for farmers to implement. This results in the potential for the farmer revert to a simpler or less cost intensive practice of simply flushing the liquid manure from the dairy operations to their lagoon.

The risk of reversion or abandonment of these practices is a very real risk in the US Dairy market.

Additionally, given the highly volatile nature of farming, both due to environmental and sociopolitical pressures, any farm could potentially risk reversion or abandonment on any

given day. The costs associated with the operational changes necessary to participate in this program are rarely, if ever, reflected in a premium price for the product. Without continued incentives or compensation, many of these practices pose a challenge to the farm's finances long term, threatening their continuity. This lack of compensation creates the potential for any farmer revert to simpler or less cost intensive anaerobic management practices used prior to the project.

Given both the large-scale financial risk (some of the technologies in question can cost hundreds of thousands of dollars) and the ever-changing bottom-line viability of the farm, it is incumbent on each farmer to make regular and deliberate decisions to continue with the intervention activity and actively maintain the resultant reductions in emissions. Dairy market price volatility as well as environmental threats to dairy farm profitability have been increasingly concerning. Sources such as the USDA and other leading publications note slow growth after periods of extreme volatility over the last few years. Commodity feed prices, replacement heifers, trucking, and more all also continue to increase in cost. These pressures have the potential to affect every farm on a moment's notice- a fire knocks out their feed supply (like the nearly annual major wildfires that occur in California) or a snowstorm decimates their herd (Midwest snowstorms in 2019 collapsed barns and froze large portions of herds to death), or a company the farm relied on for their operational continuity goes out of business. Since the volatility of commodity prices must be managed with the very real threats to operational continuity that these farmers deal with on a day-to-day basis, the average producer runs a great risk to the bottom-line of their farm any time they engage in a practice that does not have clear cut economic returns. Since the practices listed in Section 2 do not have these clear economic returns, every farmer who engages in the practices of this program, runs that risk of abandonment and reversion as long as they are self-funding these practices.

In providing additional funding through the sale of impact units, farmers are incentivized to not just to implement but to also maintain alternative manure management practices in the long term. This income stream allows the producer to be able to maintain their equipment and staff in such a way that reverting to anaerobic manure management is no longer the sounder business decision, thereby ensuring that the GHG reductions from the alternative manure management practices continue.

3 Eligibility

The activities associated with this intervention are the feeding of Bovaer daily to lactating dairy cows and the implementation of a manure management practice that diverts volatile solids from an anaerobic storage condition to a non-anaerobic one. For all producers participating in the program business-as-usual is defined as most of the farm's manure being managed under anaerobic conditions. On most concentrated dairy operations, this takes the form of a manure lagoon, either uncovered or covered, in which long term storage of liquid manure encourages generation of high levels of methane and nitrous oxide production. Intervention activities, conversely, create non-anaerobic environments for manure storage and breakdown either through the removal of water content prior to storage or storage in an environment that encourages drying of the manure, which leads to carbon dioxide production rather than methane and nitrous oxide.

To determine a producer's eligibility to participate in this intervention, the following data will be collected by the end of verification:

1. Animals included in the project located on a dairy operation
2. Animal types included; lactating and dry cattle and heifers
3. Evidence of the installation or implementation (in the case of non-technology based interventions) and operation of a non-digester manure management practices or technologies that avoid the anaerobic decomposition of manure volatile solids.
4. Evidence that the baseline manure management practices include the anaerobic decomposition of manure volatile solids stored in a lagoon or other predominantly liquid anaerobic environment.
5. Evidence the dairy farm is located in the US or US Tribal Lands
6. Completed legal attestation of voluntary compliance.
7. Completed legal attestation that the project activities do not cause material violations of applicable laws (e.g. water quality, safety, etc.).

To determine a producer's ongoing eligibility to participate in this intervention, the verifiers will track and monitor the following:

1. Number of cattle for each manure management system used on farm.
2. The diet of the cattle for each cattle type.
3. Frequency of cleaning of any holding areas for manure
4. Utility information pertaining to the management of manure on the farm.

All projects participating in this intervention must also meet the eligibility criteria outlined in Table 3.1.

Table 3.1 Summary of eligibility criteria for the generation of credits

Eligibility Criteria Summary	
Directions	Feed according to label instructions
Dose	60-80 mg 3-NOP per kg total ration DM (on-label dose as of May 2024)
Animal Types	Lactating dairy cattle
Location	U.S. or U.S. tribal lands
Reporting Period	No minimum reporting period. Maximum reporting period is 12 months. After 12 months a project may continue but it must use the most recent version of this protocol
Compliance	Attestation of voluntary compliance is required; Bovaer implementation must not be legally required.

3.1 Voluntary Compliance & Performance Standard

Projects must demonstrate a scenario that is "better than business-as-usual." Each producer whose dairy operation is included in the project must sign an attestation of voluntary compliance. Attestations must be signed prior to the commencement of verification activities each time the project is verified.

3.2 Project Start Date

The implementation start date for this intervention January 1, 2025 or the first active use date of the manure management activity concurrently with the feeding of Bovaer, whichever is later. An intervention is considered in active use on the date at which the system begins to function at the intended manure intake levels upon completion of an initial start-up period. An initial start-up period must not exceed nine months. Intended manure intake levels are defined as the planned maximum manure treatment capacity of the project activity. Projects may be submitted any time after their official start date until the end of the calendar year in which they started.

3.3 Reporting Period

The preferred monitoring period is at least one calendar month and the preferred project duration is at least 12 months. Producers do have the option of choosing a quarterly monitoring period if that best fits the needs of their business, in consultation with Athian and the verifier assigned to their intervention. After 12 months using this protocol, a project may continue, but it must use the most recent version of this protocol.

3.4 Location

Only projects located in the U.S., or on U.S. tribal lands, are eligible to generate credits under this program.

4 GHG Assessment Boundary

Due to the studies referenced below, the separation of Bovaer and AMMP SSR's as reviewed in the independent third-party validation audits of both programs is valid and does not present an overlapping risk of double counting any reductions, nor do the intervention activities affect the reductions of each other.

4.1 AMMP Emission Boundaries

SSR	GHG	Included or Excluded	Justification
Feed Cultivation	CO ₂ , N ₂ O	Excluded	Emissions from the transportation, production, and harvesting of cattle feed do not change between the baseline and project scenario.
Manure Management	CH ₄ , N ₂ O	Included	Emissions from manure are reduced by the practices included in this protocol.
Fuel & Electricity Use	CO ₂ , CH ₄ , N ₂ O	Included	Emissions from energy use for manure management (e.g., for pumps, scrapers, vehicles, heaters, etc.) may be increased or decreased by the practices included in this protocol. Electricity emissions are CO ₂ only.
Waste Processing	CO ₂ , CH ₄ , N ₂ O	Excluded	Emissions from the management of dead animals do not change between the baseline and project scenario.
Direct Land Use	CO ₂	Excluded	Emissions from land use do not change between the baseline and project scenario.
Enteric Fermentation	CH ₄	Excluded	Emissions from enteric fermentation in cattle do not change between the baseline and project scenario.

While manure management systems themselves have no effect on enteric emissions, several of the factors that go into the AMMP quantification may. Dry matter intake (DMI) and manure management system were identified during sensitivity analysis as the two primary factors affecting AMMP emissions reductions outcomes. DMI also has an outsized effect on enteric emissions, and by extension, Bovaer emissions outcomes. As part of the validation process for Alternative Manure Management, an advisory was included that cautions participating producers against leveraging dry matter intake (DMI) too heavily for milk yield increases. In the case of concurrent enrollment, this should also be considered as an advisory for Bovaer participation.

4.2 Bovaer Emission Boundaries

SSR	GHG	Included or Excluded	Justification
Direct Land Use	CO ₂	Excluded	Emissions from land use do not change between the baseline and project scenario.
Enteric Fermentation	CH ₄	Included	CH ₄ emissions from enteric fermentation are the major source of emissions in the project scenario (Kebreab et al., 2023)
Fuel & Electricity Use	CO ₂ , CH ₄ , N ₂ O	Excluded	Fossil fuel emissions from electricity and stationary fuel use do not change between the baseline and project scenario
Ingredient Production & Transport	CO ₂ , N ₂ O	Included	Emissions from the manufacturing and transport of the feed ingredient intervention are included in this program.
Manure Management	CH ₄ , N ₂ O	Excluded	Emissions from the management of manure do not change between the baseline and project scenario.
Waste Processing	CO ₂ , CH ₄ , N ₂ O	Excluded	Emissions from the management of dead animals do not change between the baseline and project scenario.

As part of the validation audit of Bovaer, the justification of its selective SSR's to just enteric methane was evaluated by SustainCert. The boundaries evaluated determined that Bovaer does not significantly affect manure emissions. The argument presented focused on five features of Bovaer and its digestion:

- Bovaer is metabolized by a unique mode of action and is reliably metabolized in the rumen.

Duin et al. demonstrated that 3-NOP is metabolized by its own mode of action within the rumen by targeting the active site of methyl-coenzyme M reductase (MCR) and is "highly specific toward methanogenic archaea." (Duin et al., 2016). This paper establishes that 3-NOP is metabolized in the rumen and is unlikely to reach the lower digestive tract of the animal.

- Bovaer is not excreted into the environment in manure, urine, or milk.

Additionally, Bampidis et al. demonstrated that 3-NOP is not excreted into the environment, supporting the Duin et al. assertion regarding its digestion. In a series of experiments, Bampidis et al. assessed radioactively marked 3-NOP concentrations in urine, feces, and milk in ruminants and concluded that 3-NOP and its degradation products are almost completely absorbed and what is not absorbed is used in the formation of endogenous compounds and exhaled CO₂. (Bampidis et al., 2021).

- Bovaer does not affect manure composition.

In 2019, Nkema et al. assessed the effects of 3-NOP on downstream manure composition characteristics and GHG emissions of 3-NOP fed and 3-NOP inoculated manure on anaerobic digestion for biogas. This study found that there were not significant differences in manure composition of 3-NOP manure and control manure.

For AMMP purposes, this also extended to the percentages of total solids and percentages of volatile solids in the manure, which are a primary driving force behind AMMP emissions. Additionally, this same study concluded that there was not a significant difference in methane yields of 3-NOP manure compared to control manure, leading to the conclusion that 3-NOP manure in an AMMP system would degrade similarly to manure from non-treated animals within that same system.

- Bovaer does not affect emissions from manure degradation after excretion.

Owens et al. found in 2020 that GHG emissions of CH₄, CO₂, NH₃, and N₂O were not fundamentally different between 3-NOP manure and control manure as the manure degraded over time. This study does also confirm the sensitivity analysis assertion that manure management systems have a bigger effect on the manure emissions profile than feed additives present. Owens also observed that many of these emissions would be most influenced by microbial activity, instead of the presence of 3-NOP or its degradation products.

- Bovaer does not affect soil health or characteristics when manure from animals consuming Bovaer is applied to cropland.

Additionally, in a subsequent study, Owens et al. found in 2021 that 3-NOP manure did not affect the soil profile when applied to crops. This study evaluated chemical, physical, and hydraulic soil health properties. While it did show some variability in protein and active C in the soil with 3-NOP manure, the study as a whole concluded that 3-NOP manure did not significantly differ in its effect when applied to soil compared to manure from non-treated cattle. AMMP does include a direct land application option for manure (daily spread), but does not otherwise quantify land use emissions of manure and for this reason, we do not believe that any changes in soil properties listed above would affect the AMMP reductions quantified on 3-NOP manure.

5 GHG Quantification

$$\Delta GHG = \Delta GHG_{AMMP} + \Delta GHG_{Bovaer} \quad (\text{Equation 5.1})$$

Where:

ΔGHG	=	Net GHG emissions from the project after implementation of the practices (kg CO ₂ e)
ΔGHG_{AMMP}	=	Net GHG emissions reductions due to alternative manure management practices (kg CO ₂ e)
ΔGHG_{Bovaer}	=	Net GHG emissions reductions due to Bovaer Reductions (kg CO ₂ e)

Table 5.1 Cattle Types

Cattle Types
Dry dairy cows
Lactating dairy cows
Replacement dairy heifers, birth to 12 months
Replacement dairy heifers, 12.1 months to calving
Beef animals ¹

Alternative Manure Management GHG Emissions Reduction is calculated as:

$$\Delta GHG_{AMMP} = \sum_T (GHG_{AMMP,B}^T - GHG_{AMMP,P}^T) \quad (\text{Equation 5.2})$$

Where:

$GHG_{AMMP,B}^T$	=	Baseline GHG emissions from manure management for cattle type T during the reporting period (kg CO ₂ e). This value is represented by the results of Equation 5.2 in Section 5. Alternative Manure Management Quantification Methodology.
$GHG_{AMMP,P}^T$	=	Project GHG emissions from manure management for cattle type T during the reporting period (kg CO ₂ e). This value is represented by the results of Equation 5.22 in Section 5. Alternative Manure Management Quantification Methodology

Bovaer GHG Emissions Reduction is calculated as:

$$\Delta GHG_{Bovaer} = \sum_T (GHG_{Bovaer,B}^T - GHG_{Bovaer,P}^T) \quad (\text{Equation 5.3})$$

Where:

$GHG_{Bovaer,B}^T$	=	Baseline GHG emissions from enteric emissions for cattle type T during the reporting period (kg CO ₂ e). This value is represented by the results of Equation 6.1 in Section 6. Bovaer Quantification Methodology for lactating cattle. For all other cattle types, this value is 0.
$GHG_{Bovaer,P}^T$	=	Project GHG emissions from enteric methane for cattle type T during the reporting period (kg CO ₂ e). This value is represented by the results of Equation 6.3 in Section 6. Bovaer Quantification Methodology for lactating cattle. For all other cattle types, this value is 0.

5.1 Alternative Manure Management Quantification

5.1.1 Baseline GHG Emissions

The baseline scenario is the situation where, in the absence of the intervention activity, animal manure is left to degrade in a predominantly anaerobic environment and the resultant CH₄ and N₂O is emitted to the atmosphere. More simply, the baseline scenario is the manure management system used on the farm in the absence of alternative manure management practices, in the average environmental conditions of the baseline period. In the baseline scenario, the retention time of manure waste in the anaerobic treatment system is greater

¹ Non-replacement calves raised for beef purposes retained on the dairy operation

than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1m (CDM AMS-III.d).

Baseline electricity and fuel emissions are averages based on sources within the GHG assessment boundary on data from the 24 months prior to the beginning of the project. This baseline is based on methodologies from the U.S. Livestock Project Protocol and the CDM AMS-III.d Small-Scale Methodology: Methane recovery in animal manure systems program.

Baseline manure emissions represent the emissions that would have occurred with the previous manure management system (i.e. business-as-usual, the manure management that would have occurred in the absence of the project) under the conditions of the baseline period. This baseline is established on averages of data from the 24 months prior to the implementation start date of the intervention.

$$GHG_{AMMP,B}^T = GHG_{man,B}^T + GHG_{elec,B}^T + GHG_{fuel,B}^T \quad (\text{Equation 5.2})$$

Where:

$GHG_{man,B}^T$	=	Baseline GHG emissions from manure for cattle type T during the reporting period (kg CO ₂ e) that would have occurred in the absence of the project
$GHG_{elec,B}^T$	=	Baseline GHG emissions from electricity use for manure management activities for cattle type T during the reporting period (kg CO ₂ e) that would have occurred in the absence of the project
$GHG_{fuel,B}^T$	=	Baseline GHG emissions from fuel use for manure management for cattle type T during the reporting period (kg CO ₂ e) that would have occurred in the absence of the project

5.1.1.1 Baseline GHG Emissions from Manure Management

Baseline GHG emissions from manure production and management are calculated according to Equations 5.3 to 5.20:

$$GHG_{man,B}^T = \sum_T ((CH_4^T_{man,B} \times GWP_{CH_4}) + (N_2O^T_{man,B} \times GWP_{N_2O})) \quad (\text{Equation 5.3})$$

Where:

$CH_4^T_{man,B}$	=	Baseline methane emissions from manure for cattle type T during the reporting period that would have occurred in the absence of the project (kg CH ₄)
GWP_{CH_4}	=	Global warming potential of methane (kgCO ₂ e kgCH ₄ ⁻¹)
$N_2O^T_{man,B}$	=	Baseline nitrous oxide emissions from manure production for cattle type T during the reporting period that would have occurred in the absence of the project (kg N ₂ O)
GWP_{N_2O}	=	Global warming potential of nitrous oxide (kg CO ₂ e kg N ₂ O ⁻¹)

5.1.1.1.1 Baseline Manure Methane Emissions

Baseline methane emissions from manure management are the sum of methane emissions from the predominantly anaerobic (liquid) storage/treatment systems and predominantly non-anaerobic (dry/solid) storage/treatment systems that would have occurred in the absence of

the project. Manure methane emissions are calculated using a Tier 2 approach consistent with both IPCC and Environmental Protection Agency (EPA) quantification methodologies. This protocol also leverages the approaches used by the Climate Action Reserve's U.S. Livestock Project Protocol, the California Air Resources Board's Compliance Offset Protocol for capturing and destroying methane from manure management systems, and the quantification methodology for California Department of Food and Agriculture's Alternative Manure Management Program (Climate Action Reserve, 2013; California Air Resources Board, 2014, 2023).

$$CH_4^T_{man,B} = CH_4^T_{liq,B} + CH_4^T_{dry,B} \quad (\text{Equation 5.4})$$

Where:

$CH_4^T_{liq,B}$	=	Baseline methane emissions from manure for cattle type T in predominantly anaerobic (liquid) manure storage/treatment that would have occurred during the reporting period in the absence of the project (kg CH ₄)
$CH_4^T_{dry,B}$	=	Baseline methane emissions from manure for cattle type T in predominantly non-anaerobic (dry/solid) manure storage/treatment that would have occurred during the reporting period in the absence of the project (kg CH ₄)

5.1.1.1.2 Baseline Manure Methane Emissions from Anaerobic Storage/Treatment

$$CH_4^T_{liq,B} = \sum_{AS,T} (VS_{deg,AS,B}^T \times B_0^T) \times \rho_{CH_4} \quad (\text{Equation 5.5})$$

Where:

$VS_{deg,AS,B}^T$	=	Volatile solids from cattle type T that would have degraded during the reporting period in anaerobic storage/treatment system AS in the absence of the project (kg dry matter)
B_0^T	=	Maximum methane-producing capacity of manure for cattle type T (see Table 5.2, m ³ CH ₄ kg VS ⁻¹)
ρ_{CH_4}	=	Density of methane = 0.67 (1 atm, 60°F) (kg m ⁻³) ²

With:

$$VS_{deg,AS,B}^T = f_i \times VS_{avail,AS,B,i}^T \quad (\text{Equation 5.6})$$

Where:

f_i	=	The van't Hoff-Arrhenius factor for the current reporting period i
$VS_{avail,AS,B,i}^T$	=	Volatile solids from cattle type T that would have been available for degradation during the current reporting period i in anaerobic storage/treatment system AS in the absence of the project (kg dry matter)

² Taken from Equation 5-26 of Powers et al. 2014

With:

$$f_i = MIN\left(\exp\left[\frac{E(T_2 - T_1)}{R T_1 T_2}\right], 0.95\right) \quad (\text{Equation 5.7})$$

Where:

E	=	Activation energy constant (15,175 cal/mol)
T_1	=	303.16 Kelvin
T_2	=	Average ambient temperature (K) in the reporting period ($K = ^\circ C + 273$). If $T_2 < 5^\circ C$, then $f = 0.104$.
R	=	Ideal gas constant (1.987 cal/K-mol)

And:

$$VS_{avail,AS,B,i}^T = (VS_i^T \times C_i^T \times PS_{AS,B,i}^T \times t_i \times 0.8) + (VS_{avail,AS,B,i-1}^T - VS_{deg,AS,B,i-1}^T) \quad (\text{Equation 5.8})^3$$

Where:

VS_i^T	=	Volatile solids produced by cattle type T on a dry matter basis during the reporting period i (kg cattle ⁻¹ day ⁻¹)
C_i^T	=	Average number of animals for each cattle type T during the reporting period i
$PS_{AS,B,i}^T$	=	Percent of manure solids that would have been sent to (managed in) anaerobic manure storage/treatment system AS from cattle type T during the reporting period i in the absence of the project. Dairy-specific data must be used. ⁴
t_i	=	Number of days in the reporting period i
0.8	=	System calibration factor ⁵
$VS_{avail,AS,B,i-1}^T$	=	Volatile solids from cattle type T from the previous reporting period $i - 1$ that would have been available for degradation in anaerobic storage/treatment system AS (kg dry matter) in the absence of the project. ⁶
$VS_{deg,AS,B,i-1}^T$	=	Volatile solids from cattle type T that would have degraded during the previous reporting period $i - 1$ in anaerobic storage/treatment system AS in the absence of the project (kg dry matter)

³ Equation from CDFA AMMP, page 13

⁴ The value of $PS_{AS,B,i}^T$ must take into account: the fraction of volatile solids that would have been recoverable (i.e., able to be collected for transfer into the baseline anaerobic storage/treatment system) in the absence of the project, which is a function of the amount of time animals would have spent in different enclosure types; the type and efficiency the baseline manure collection system; and the efficiency of the baseline solid separation technology, if any.

⁵ From California Air Resources Board (2023) and US EPA (2016).

⁶ For the first reporting period of the project, if there are no site-specific data on the mass of volatile solids in the anaerobic storage/treatment system at the beginning of the project, the value may be calculated in one of two ways. If there are monthly data on herd characteristics, animal diet, and average site temperature since the last cleanout of the anaerobic storage/treatment system, the amount of volatile solids at the beginning of the project shall be calculated by setting $(VS_{avail,AS,B,i-1}^T - VS_{deg,AS,B,i-1}^T)$ to zero in the month following the cleanout and using the equations in this protocol to calculate the buildup of volatile solids in the months since the cleanout. Otherwise, $(VS_{avail,AS,B,i-1}^T - VS_{deg,AS,B,i-1}^T)$ will be set to zero in the first reporting period. If the retention time for volatile solids in the anaerobic storage/treatment system is less than or equal to 30 days, set $(VS_{avail,AS,B,i-1}^T - VS_{deg,AS,B,i-1}^T)$ to zero. Similarly, for any reporting period following the complete drainage and cleaning of solid buildup from the anaerobic storage/treatment system, $(VS_{avail,AS,B,i-1}^T - VS_{deg,AS,B,i-1}^T)$ must be set to zero.

$$VS_i^T = [DMI^T \times (1 - DE^T) + (UE^T \times DMI^T)] \times (1 - ASH^T) \quad (\text{Equation 5.9})$$

Where:

DMI^T	=	Average dry matter intake per animal of cattle type T (kg head ⁻¹ day ⁻¹) during the reporting period. Calculated based on known intake (feedyard) or estimated intake equations (other cattle categories)
DE^T	=	Diet digestibility of the feed for cattle type T in the reporting period as a fraction of gross energy (see Table 5.3) (dimensionless) Where DE is not available from primary data sources, projects may choose to use the Athian Dairy Total Digestible Nutrients Tool instead. Methodology for this tool is available as Appendix D. Athian Dairy Total Digestible Nutrients Tool Methodology .
UE^T	=	Urinary energy of cattle type T in the reporting period expressed as a fraction of gross energy (dimensionless). A default value of 4% is used for UE. This value can be reduced to 2% for cattle fed with 85 percent or more grain in their diet.
ASH^T	=	Ash content of feed for cattle type T in the reporting period (see Table 5.3) (%)

DE and Ash are calculated as a weighted average using the data from **Table 5.3**. Producers may provide alternative approaches to determine DE and Ash content of feed.

Table 5.2 Maximum Methane-producing Capacity of Manure for Cattle⁷

Cattle Types (T)	B_o^T (m ³ CH ₄ kg ⁻¹ VS added)
Dairy Cattle (both dry and lactating)	0.24
Replacement Heifers	0.17
Dairy Beef Steers	0.33

Table 5.3 Examples of DE and Ash Content by Feed Type⁸

Feed	DE (% of GE)	Ash %
Alfalfa hay early bloom	63.72	8
Alfalfa silage	60.71	9
Corn grain, whole	88.85	2
Corn silage, mature well eared	72.88	5
DDGS, dry mill	76.88	4
Distillers grain, corn with solubles	81.50	5
Grass hay	N/A	6
Grass silage	N/A	8
Oat grain	75.63	4

⁷ Table from Table 5-19 of Powers et al. 2014.

⁸ Taken from Appendix 5-B of Powers et al. 2014. Additional feed values are available in the Powers Appendix.

Soybean hulls	66.86	5
Soybean meal, solv. ext. 44% CP	79.50	7
Winter wheat grain	86.45	2
Canola meal, solv. ext	N/A	8
Cottonseed, whole	N/A	5
Citrus pulp dried	N/A	7
Wheat midds	N/A	5

5.1.1.1.3 Baseline Manure Methane Emissions from Non-Anaerobic Storage/Treatment

$$CH_4^{dry,B} = \sum_{S,T} (C_i^T \times VS_i^T \times PS_{S,B,i}^T \times MCF_S \times B_0^T \times \rho_{CH_4} \times t) \quad (\text{Equation 5.10})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
VS^T	=	Volatile solids produced by cattle type T on a dry matter basis during the reporting period (kg cattle ⁻¹ day ⁻¹)
$PS_{S,B,i}^T$	=	Percent of manure solids that would have been sent to (managed in) non-anaerobic manure storage/treatment system S from cattle type T during the reporting period in the absence of the project. Dairy-specific data must be used. ⁹
MCF_S	=	Methane conversion factor for non-anaerobic manure storage/treatment system S during the reporting period, based on T_2 , in Table 5.4 (%)
B_0^T	=	Maximum methane-producing capacity of manure for cattle type T in Table 5.2 (m ³ CH ₄ kg VS ⁻¹)
ρ_{CH_4}	=	Density of methane = 0.67 (1 atm, 60°F) (kg m ⁻³)
t	=	Number of days in the reporting period

$$VS_i^T = [DMI^T \times (1 - DE^T) + (UE^T \times DMI^T)] \times (1 - ASH^T) \quad (\text{Equation 5.11})$$

Where:

DMI^T	=	Dry matter intake of cattle type T in the reporting period (kg)
DE^T	=	Diet digestibility of the feed for cattle type T in the reporting period as a fraction of gross energy (see Table 5.3) (%) Where DE is not available from primary data sources, projects may choose to use the Athian Dairy Total Digestible Nutrients Tool instead. Methodology for this tool is available as Appendix D. Athian Dairy Total Digestible Nutrients Tool Methodology .

⁹ The value of PSS,BT must take into account the fraction of volatile solids that would have been deposited on land (e.g., pasture, open lot) and the fraction that would have been recoverable (i.e., able to be collected for transfer into a non-anaerobic storage/treatment system, if any) in the absence of the project. These fractions are functions of the amount of time animals would have spent on pasture vs. different enclosure types in the absence of the project. PSS,BT also depends on the type and efficiency of the baseline manure collection system in collecting recoverable manure and the efficiency of the baseline solid separation technology, if any.

UE^T	=	Urinary energy of cattle type T in the reporting period expressed as a fraction of gross energy (dimensionless). A default value of 4% is used for UE. This value can be reduced to 2% for cattle fed with 85 percent or more grain in their diet.
ASH^T	=	Ash content of feed for cattle type T in the reporting period (see Table 5.3) (dimensionless)

Table 5.4 Methane Conversion Factor (MCF) for Non-Anaerobic Storage / Treatment Systems.¹⁰
System definitions are in Table 5.5.

System	Average Temperature in Reporting Period, T2 (°C)									
	<10	≤11	≤12	≤13	≤14	≤15	≤16	≤17	≤18	≤19
Pasture/Range/ Paddock	1%	1%	1%	1%	1%	1.5%	1.5%	1.5%	1.5%	1.5%
Daily Spread	0.1%	0.1%	0.1%	0.1%	0.1%	0.5%	0.5%	0.5%	0.5%	0.5%
Solid Storage	2%	2%	2%	2%	2%	4%	4%	4%	4%	4%
Dry Lot	1%	1%	1%	1%	1%	1.5%	1.5%	1.5%	1.5%	1.5%
Liquid/Slurry (With Natural Crust Cover)	10%	11%	13%	14%	15%	17%	18%	20%	22%	24%
Liquid/Slurry (Without Natural Crust Cover)	17%	19%	20%	22%	25%	27%	29%	32%	35%	39%
Pit Storage (<1 Month)	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%
Pit Storage (>1 Month)	17%	19%	20%	22%	25%	27%	29%	32%	35%	39%
Deep Bedding (<1 Month)	3%	3%	3%	3%	3%	3%	3%	3%	3%	3%
Deep Bedding (>1 Month)	17%	19%	20%	22%	25%	27%	29%	32%	35%	39%
Composting -- In-Vessel Or Aerated Static Pile	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%
Composting -- Windrows	0.5%	0.5%	0.5%	0.5%	0.5%	1%	1%	1%	1%	1%
Weeping Wall	22%	22%	22%	22%	22%	22%	22%	22%	22%	22%

¹⁰ Values from CDFA (2023) AMMP Draft Benefits Calculator Tool. The monthly average ambient temperature for the reporting period must be obtained from the closest weather station with available data; if applicable, the weather station should be located in the same air basin.

Table 5.4 Methane Conversion Factor (MCF) for Non-Anaerobic Storage / Treatment Systems, Continued. Note: Any monthly average greater than 28°C should utilize the MCF listed for 28°C

System	Monthly Average Temperature Range (°C)								
	<20	≤21	≤22	≤23	≤24	≤25	≤26	≤27	≤28
Pasture/Range/ Paddock	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	2%	2%	2%
Daily Spread	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	1%	1%	1%
Solid Storage	4%	4%	4%	4%	4%	4%	5%	5%	5%
Dry Lot	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%	2%	2%	2%
Liquid/Slurry (With Natural Crust Cover)	26%	29%	31%	34%	37%	41%	44%	48%	50%
Liquid/Slurry (Without Natural Crust Cover)	42%	46%	50%	55%	60%	65%	71%	78%	80%
Pit Storage (<1 Month)	3%	3%	3%	3%	3%	3%	3%	3%	3%
Pit Storage (>1 Month)	42%	46%	50%	55%	60%	65%	71%	78%	80%
Deep Bedding (<1 Month)	3%	3%	3%	3%	3%	3%	3%	3%	3%
Deep Bedding (>1 Month)	42%	46%	50%	55%	60%	65%	71%	78%	90%
Composting -- In-Vessel Or Aerated Static Pile	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%	0.5%
Composting -- Windrows	1%	1%	1%	1%	1%	1%	1.5%	1.5%	1.5%
Weeping Wall	22%	22%	22%	22%	22%	22%	22%	22%	22%

5.1.1.1.4 Baseline Manure Nitrous Oxide Emissions

Baseline emissions of N₂O are calculated as follows:

$$N_2O_{man,B}^T = \sum_T (N_2O_{dir,B}^T + N_2O_{ind,l,B}^T + N_2O_{ind,v,B}^T) \quad (\text{Equation 5.12})$$

Where:

$N_2O_{dir,B}^T$	=	Direct N ₂ O emissions from management of manure from cattle type T that would have occurred in the absence of the project (kg N ₂ O)
$N_2O_{ind,l,B}^T$	=	Indirect N ₂ O emissions from leaching of manure from cattle type T that would have occurred in the absence of the project (kg N ₂ O)
$N_2O_{ind,v,B}^T$	=	Indirect N ₂ O emissions from cattle type T from the volatilization of NH ₃ and NO _x that would have occurred in the absence of the project (kg N ₂ O)

5.1.1.1.5 Baseline Direct N₂O Emissions from Manure Management

The calculation of direct N₂O emissions entails multiplying the total amount of nitrogen excretion in each type of manure management system by an emission factor for that type of system and summing the obtained values. The emissions shall then be calculated using Equation 5.13:

$$N_2O_{dir,B}^T = (\sum_S (\sum_T (C^T \times N_{ex}^T \times AWMS_{S,TB}))) \times EF_{dir,S} \times \frac{44}{28} \quad (\text{Equation 5.13})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
$EF_{dir,S}$	=	Emission factor for direct N ₂ O emissions from non-anaerobic manure management system S (dimensionless) in Table 5.5.
$\frac{44}{28}$	=	Conversion factor between N ₂ O–N emissions and N ₂ O emissions
N_{ex}^T	=	Average N excretion per animal of cattle type T (kg head ⁻¹)
$AWMS_{S,TB}$	=	Baseline fraction of total annual volatile solids for each livestock category T that is managed in manure management system S.

Table 5.5 Default Emission Factors for Direct N₂O Emissions from Manure Management¹¹

System	Definition	$EF_{dir,S}$
Pasture/ Range/ Paddock	The manure from pasture and range grazing animals is allowed to lie as-is and is not managed.	Direct and indirect N ₂ O emissions associated with the manure deposited on agricultural soils and pasture, range, and paddock systems are not included in this protocol.
Daily Spread	Manure is routinely removed from a confinement facility and is applied to cropland or pasture within 24 hours of excretion. N ₂ O emissions during storage and treatment are assumed to be zero.	0
Solid Storage	The storage of manure, typically for a period of several months, in unconfined piles or stacks. Manure is able to be stacked due to the presence of a sufficient amount of bedding material or loss of moisture by evaporation.	0.010
Solid Storage – Covered/Compacted	Similar to solid storage, but the manure pile is a) covered with a plastic sheet to reduce the surface of manure exposed to air and/or b) compacted to increase the density and reduce the free air space within the material.	0.01

¹¹ Adapted from Table 10.21 in Chapter 10 of IPCC (2019).

Solid Storage – Bulking Agent Addition	Specific materials (bulking agents) are mixed with the manure to provide structural support. This allows the natural aeration of the pile, thus enhancing decomposition. (e.g. sawdust, straw, coffee husks, maize stover)		0.005
Solid Storage – Additives	The addition of specific substances to the pile in order to reduce gaseous emissions. Addition of certain compounds such as attapulgate, dicyandiamide or mature compost have shown to reduce N ₂ O emissions; while phosphogypsum reduce CH ₄ emission		0.005
Dry Lot	A paved or unpaved open confinement area without any significant vegetative cover where accumulating manure may be removed periodically. Dry lots are most typically found in dry climates but also are used in humid climates.		0.02
Liquid/Slurry	Manure is stored as excreted or with some minimal addition of water to facilitate handling and is stored in either tanks or earthen ponds.	With natural crust cover	0.005
		Without natural crust cover	0
		Cover	0.005
Uncovered Anaerobic Lagoon	Anaerobic lagoons are designed and operated to combine waste stabilization and storage. Lagoon supernatant is usually used to remove manure from the associated confinement facilities to the lagoon. Anaerobic lagoons are designed with varying lengths of storage (up to a year or greater), depending on the climate region, the volatile solids loading rate, and other operational factors. The water from the lagoon may be recycled as flush water or used to irrigate and fertilize fields.		0
Pit Storage Below Animal Confinements	Collection and storage of manure usually with little or no added water typically below a slatted floor in an enclosed animal confinement facility.		0.002
Anaerobic Digester	Anaerobic digesters are designed and operated for waste stabilization by the microbial reduction of complex organic compounds to CH ₄ and CO ₂ , which is captured and flared or used as a fuel.		0.0006
Deep Bedding	As manure accumulates, bedding is continually added to absorb moisture over a production cycle and possibly for as long as 6 to 12 months. This manure management system also is known as a bedded pack manure management system and may be combined with a dry lot or pasture.	No mixing	0.01
		Active mixing	0.07
Composting – In-Vessel	Composting, typically in an enclosed channel, with forced aeration and continuous mixing.		0.006
Composting – Static Pile (Forced Aeration)	Composting in piles with forced aeration but no mixing.		0.010
Composting – Intensive Windrow (Frequent Turning)	Composting in windrows with regular turning for mixing and aeration.		0.005

Composting – Passive Windrow (Infrequent Turning)	Composting in windrows with infrequent turning for mixing and aeration.		0.005
Aerobic Treatment	The biological oxidation of manure collected as a liquid with either forced or natural aeration. Natural aeration is limited to aerobic and facultative ponds and wetland systems and is due primarily to photosynthesis. Hence, these systems typically become anoxic during periods without sunlight.	Natural aeration systems	0.01
		Forced aeration systems	0.005

Rates of N excretion are calculated using Equations 5.14 and 5.15:

$$N_{ex}^T = N_{in}^T \times (1 - N_{ret}^T) \times t \quad (\text{Equation 5.14})$$

$$N_{in}^T = DMI_{per}^T \times \frac{CP^T}{6.25} \quad (\text{Equation 5.15})$$

Where:

N_{in}^T	=	Average daily nitrogen intake per animal of cattle type T in the reporting period (kg head ⁻¹ day ⁻¹)
N_{ret}^T	=	Fraction of daily N intake that is retained by cattle type T (dimensionless). Calculated according to (Equation 5.16).
DMI_{per}^T	=	Average dry matter intake per animal of cattle type T (MJ head ⁻¹ day ⁻¹) during the reporting period. Calculated based on known intake (feedyard) or estimated intake equations (other cattle categories)
CP^T	=	Crude protein content in the overall diet for cattle type T in the reporting period (%)
t	=	Number of days in the reporting period

The fraction of nitrogen retained by the animal is calculated according to Equation 5.16:

$$N_{ret}^T = \frac{M \times \left(\frac{M_p}{100}\right)}{6.38} + \frac{WG^T \times \frac{268 - \frac{7.03 \times NE_g}{WG}}{1000}}{6.25} \quad (\text{Equation 5.16})$$

$$NE_g = 22.02 \times \left(\frac{BW^T}{\gamma \times MW^T}\right)^{0.75} \times (WG^T)^{1.097} \quad (\text{Equation 5.17})$$

Where:

M	=	Average milk production per cow during the reporting period (kg head ⁻¹ day ⁻¹); this is 0 for other cattle types
M_p	=	Percentage of protein in the milk during the reporting period (%)
6.38	=	Conversion factor from milk protein to milk nitrogen (dimensionless)

WG^T	=	Average weight gain of the animals in the population of cattle type T (kg day ⁻¹). Note: because lactating and dry cattle are assumed to have no weight gain or loss per Table 5.6-C and setting WG(T) to 0 would cause an error in the calculation, the second half of Equation 5.17 is set to 0 for lactating and dry cattle.
1,000	=	Conversion from g to kg
NE_G	=	Net energy for growth (MJ day ⁻¹)
268	=	Constant derived from Equation 3-8 in NRC (1996), g protein kg ⁻¹ animal ⁻¹
7.03	=	Constant derived from Equation 3-8 in NRC (1996), g protein MJ ⁻¹ animal ⁻¹
6.25	=	Conversion factor from kg of dietary crude protein to kg of dietary N (dimensionless)
BW^T	=	Average live body weight (BW) of the animals in the population of cattle type T in kg
γ	=	Coefficient (dimensionless) with a value of 0.8 for females (heifers & cows), 1.0 for castrates (steers, mixed & holsteins), and 1.2 for bulls (NRC, 1996)
MW^T	=	Mature live body weight of an adult animal of cattle type T in moderate body condition in kg, either from producer records or using defaults in Table 5.6

Table 5.6-A Mature Weights by Cattle Type

Cattle Type	MW	Note
Dairy cows	650 kg	Derived from IPCC guidance from 2019. This value applies for lactating dairy cattle, dry dairy cattle, and dairy heifers.
Dairy beef steers	820 kg	Derived from IPCC guidance from 2019.

Table 5.6-B Average Live Body Weight by Cattle Type. Mature weights listed in Table 5.6-A.

Cattle Type	BW^T	Note
Lactating Cows	650 kg	Derived from IPCC guidance from 2019
Dry Cows	650 kg	Derived from IPCC guidance from 2019
Replacement Heifers	400 kg	Derived from IPCC guidance from 2019
Dairy Beef Steers	300kg	Derived from IPCC guidance from 2019

Table 5.6-C Weight Gain Per Day by Cattle Type

Cattle Type	WG^T	Note
Lactating Cows	0 kg/day	Derived from IPCC guidance from 2019
Dry Cows	0 kg/day	Derived from IPCC guidance from 2019
Replacement Heifers	0.5 kg/day	Derived from IPCC guidance from 2019
Dairy Beef Steers	0.5 kg/day	Derived from IPCC guidance from 2019

5.1.1.1.6 Baseline Indirect N₂O Emissions from Leaching of Manure

Nitrogen is lost through runoff and leaching into soils from the solid storage of manure in outdoor areas and feedlots. The baseline amount of nitrous oxide that would have been emitted through leaching in the absence of the project shall be calculated using Equation 5.18:

$$N_2O_{ind,l,B}^T = \sum_S (\sum_T (C^T \times N_{ex}^T \times AWMS_{S,TB} \times PL_S^T)) \times EF_l \times \frac{44}{28} \quad (\text{Equation 5.18})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
N_{ex}^T	=	Average N excretion per animal of cattle type T (kg head ⁻¹)
PL_S^T	=	Percent of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in non-anaerobic manure management system S (Table 5.7) (%)
EF_l	=	Emission factor for N ₂ O emissions from nitrogen leaching and runoff. The default value is 0.011 kg N ₂ O-N (kg N leaching/runoff) ¹²
$\frac{44}{28}$	=	Conversion factor between N ₂ O-N emissions and N ₂ O emissions
$AWMS_{S,TB}$	=	Baseline fraction of total annual volatile solids for each livestock category T that is managed in manure management system S

Table 5.7 Percent of managed manure nitrogen losses due to leaching and volatilization in manure management systems.¹³ System definitions are in Table 5.5.

System		PL_S^T (%)	PV_S^T (%)
Pasture/Range/Paddock		Indirect N ₂ O emissions associated with the manure deposited on agricultural soils and pasture, range, and paddock systems are not included in this protocol.	Indirect N ₂ O emissions associated with the manure deposited on agricultural soils and pasture, range, and paddock systems are not included in this protocol.
Daily Spread		0	7
Solid Storage		2	30
Solid Storage – Covered/Compacted		0	14
Solid Storage – Bulking Agent Addition		2	38
Solid Storage – Additives		2	11
Dry Lot		3.5	30
Liquid/Slurry	With natural crust cover	0	30
	Without natural crust cover	0	48

¹² Additional emission factors may be used from Table 11.3 in Chapter 11 of IPCC (2019).

¹³ From Table 10.22 in Chapter 10 of IPCC (2019).

	Cover	0	10
Uncovered Anaerobic Lagoon		0	35
Pit Storage Below Animal Confinements		0	28
Deep Bedding	No mixing	3.5	25
	Active mixing	3.5	25
Composting – In-Vessel		0	45
Composting – Static Pile (Forced Aeration)		6	50
Composting – Intensive Windrow (Frequent Turning)		6	50
Composting – Passive Windrow (Infrequent Turning)		4	45

5.1.1.1.7 Baseline Indirect N₂O Emissions from Volatilization of NH₃ and NO_x

Nitrogen in the volatilized form of ammonia may be deposited at sites downwind from manure handling areas and contribute to indirect N₂O emissions. The baseline amount of nitrous oxide emitted through volatilization in the forms of NH₃ and NO_x shall be calculated using Equation 5.19:

$$N_2O_{ind,v,B}^T = \sum_S \left(\sum_T (C^T \times N_{ex}^T \times AWMS_{S,TB} \times PV_S^T) \right) \times EF_v \times \frac{44}{28} \quad (\text{Equation 5.19})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
N_{ex}^T	=	Average N excretion per animal of cattle type T (kg head ⁻¹)
PV_S^T	=	Percent of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x from non-anaerobic manure management system S (Table 5.7) (%)
EF_v	=	Emission factor for N ₂ O emissions from nitrogen volatilization. The default value is 0.010 kg N ₂ O-N (kg N leaching/runoff) ¹⁴
$\frac{44}{28}$	=	Conversion factor between N ₂ O-N emissions and N ₂ O emissions
$AWMS_{S,TB}$	=	Baseline fraction of total annual volatile solids for each livestock category T that is managed in manure management system S

5.1.1.2 Baseline GHG Emissions from Electricity Use

Baseline CO₂ emissions from electricity consumption are determined using averages from the 24 months prior to the adoption of alternative manure management practices. In the absence

¹⁴ Additional emission factors may be used from Table 11.3 in Chapter 11 of IPCC (2019).

of manure management-specific data (i.e., electricity consumed by equipment for activities including but not limited to moving, treating, drying, storing, and spreading manure) for each cattle type, total electricity consumption by the dairy may be used. CO₂ emissions from electricity use are calculated by multiplying the total quantity of electricity consumed and the emissions factor for electricity for the state in which the dairy is located.

$$GHG_{elec,B}^T = Elec_B^T \times EF_{elec} \times \frac{1}{2.2046} \quad (\text{Equation 5.20})$$

Where:

$GHG_{elec,B}^T$	=	Baseline CO ₂ emissions from electricity use for manure management activities for cattle type T that would have occurred during the reporting period in the absence of the project (kg CO ₂)
$Elec_B^T$	=	The average amount of electricity that would have been used for manure management activities for cattle type T in the absence of the project, based on average electricity consumption for the same time as the reporting period from the 2 years prior to the project start date ¹⁵ (MWh)
EF_{elec}	=	Emissions factor for electricity in the state in which the dairy is located (lbs CO ₂ /MWh) ¹⁶
2.2046	=	Conversion factor from lbs to kg

5.1.1.3 Baseline GHG Emissions from Fuel Use

Baseline GHG emissions from fossil fuel use are determined using averages from the 24 months prior to the adoption of alternative manure management practices. In the absence of manure management-specific data (e.g., fuel consumed by equipment for manure management, including but not limited to generators for pumps, manure drying equipment, vehicles, etc.), total fuel consumption by the dairy may be used. GHG emissions from fuel use are calculated by multiplying the total quantity of each type of fuel used and the emissions factor for that type of fuel and then summing the terms.

$$GHG_{fuel,B}^T = \sum_f (Fuel_{f,B}^T \times EF_{fuel,f}) \quad (\text{Equation 5.21})$$

Where:

$GHG_{fuel,B}^T$	=	Baseline GHG emissions from fuel use for manure management activities for cattle type T that would have occurred during the reporting period in the absence of the project (kg CO ₂ eq)
$Fuel_{f,B}^T$	=	The average amount of fuel that would have been used for manure management activities for cattle type T in the absence of the project, based on average fuel consumption for the same time as the reporting period from the 2 years prior to the project start date ¹⁷ (mmBtu or gallon)
$EF_{fuel,f}$	=	Fuel-specific emission factor (Table 5.8, kg/mmBtu or kg/gal)

¹⁵ For instance, if the current reporting period is January, the baseline will be the average electricity use of the two Januarys prior to the start of the project.

¹⁶ State-specific emissions factors are available via U.S. Energy Information Administration (2023). US Electricity Profile 2022. <https://www.eia.gov/electricity/state/>

¹⁷ For instance, if the current reporting period is January, the baseline will be the average fuel consumption of the two Januarys prior to the start of the project.

Table 5.8 Fuel-specific emissions factors¹⁸

Fuel Type	EF_f	Unit
Diesel	10.229	kg CO ₂ e / gallon
Fuel Oil	10.998	kg CO ₂ e / gallon
Kerosene	10.184	kg CO ₂ e / gallon
Propane	5.742	kg CO ₂ e / gallon
Gasoline	8.813	kg CO ₂ e / gallon
Natural Gas	53.117	kg CO ₂ e / mmBtu

5.1.2 Project GHG Emissions

Project GHG emissions for each reporting period are the sum of emissions from manure, electricity use, and fuel consumption in that reporting period, as calculated in Equation 5.22:

$$GHG_{AMMP,P}^T = GHG_{man,P}^T + GHG_{elec,P}^T + GHG_{fuel,P}^T \quad (\text{Equation 5.22})$$

Where:

$GHG_{man,P}^T$	=	Project GHG emissions from manure for cattle type T during the reporting period (kg CO ₂ e)
$GHG_{elec,P}^T$	=	Project GHG emissions from electricity use for manure management activities for cattle type T during the reporting period (kg CO ₂ e)
$GHG_{fuel,P}^T$	=	Project GHG emissions from fossil fuel use for manure management for cattle type T during the reporting period (kg CO ₂ e)

5.1.2.1 Project GHG Emissions from Manure Management

Project GHG emissions from manure production and management are calculated according to Equations 5.23 to 5.34:

$$GHG_{man,P}^T = \sum_T ((CH_4^T_{man,P} \times GWP_{CH_4}) + (N_2O^T_{man,P} \times GWP_{N_2O})) \quad (\text{Equation 5.23})$$

Where:

$CH_4^T_{man,P}$	=	Project methane emissions from manure for cattle type T during the reporting period (kg CH ₄)
GWP_{CH_4}	=	Global warming potential of methane (tCO ₂ e tCH ₄ ⁻¹)

¹⁸ Values calculated from data in U.S. Environmental Protection Agency. (2023). Emissions Factors for Greenhouse Gas Inventories. https://www.epa.gov/system/files/documents/2023-03/ghg_emission_factors_hub.pdf using $GWP_{CH_4} = 29.8$ and $GWP_{N_2O} = 273$ (Forster et al. 2021).

$N_2O_{man,P}^T$	=	Project nitrous oxide emissions from manure production for cattle type T during the reporting period (kg N ₂ O)
GWP_{N_2O}	=	Global warming potential of nitrous oxide (kg CO ₂ e kg N ₂ O ⁻¹)

5.1.2.1.1 Project Manure Methane Emissions

Project methane emissions from manure management are the sum of methane emissions from the predominantly anaerobic (liquid) storage/treatment systems and predominantly non-anaerobic (dry/solid) storage/treatment systems of the project.

$$CH_4^T_{man,P} = CH_4^T_{liq,P} + CH_4^T_{dry,P} \quad (\text{Equation 5.24})$$

Where:

$CH_4^T_{liq,P}$	=	Project methane emissions from manure for cattle type T in predominantly anaerobic (liquid) manure storage/treatment during the reporting period (kg CH ₄)
$CH_4^T_{dry,P}$	=	Project methane emissions from manure for cattle type T in predominantly non-anaerobic (dry/solid) manure storage/treatment during the reporting period (kg CH ₄)

5.1.2.1.2 Project Manure Methane Emissions from Anaerobic Storage/Treatment

$$CH_4^T_{liq,P} = \sum_{S,T} (VS_{deg,AS,P}^T \times B_0^T) \times \rho_{CH_4} \quad (\text{Equation 5.25})$$

Where:

$VS_{deg,AS,P}^T$	=	Volatile solids from cattle type T degraded during the reporting period in anaerobic project storage/treatment system AS (kg dry matter)
B_0^T	=	Maximum methane-producing capacity of manure for cattle type T (Table 5.2 Maximum Methane-producing Capacity of Manure for Cattle, m ³ CH ₄ kg VS ⁻¹)
ρ_{CH_4}	=	Density of methane = 0.67 (1 atm, 60°F) (kg m ⁻³)

With:

$$VS_{deg,AS,P}^T = f_i \times VS_{avail,AS,P,i}^T \quad (\text{Equation 5.26})$$

Where:

f_i	=	The van't Hoff-Arrhenius factor for the current reporting period i
$VS_{avail,AS,P,i}^T$	=	Volatile solids from cattle type T available for degradation during the current reporting period i in anaerobic project storage/treatment system AS (kg dry matter)

With:

$$f_i = MIN\left(\exp\left[\frac{E(T_2 - T_1)}{R T_1 T_2}\right], 0.95\right) \quad (\text{Equation 5.27})$$

Where:

E	=	Activation energy constant (15,175 cal/mol)
T_1	=	303.16 Kelvin
T_2	=	Average ambient temperature (K) in the reporting period ($K = ^\circ C + 273$). If $T_2 < 5^\circ C$, then $f = 0.104$.
R	=	Ideal gas constant (1.987 cal/K-mol)

And:

$$VS_{avail,AS,P,i}^T = (VS_i^T \times C_i^T \times PS_{AS,P,i}^T \times t_i \times 0.8) + (VS_{avail,AS,P,i-1}^T - VS_{deg,AS,P,i-1}^T) \quad (\text{Equation 5.28})$$

Where:

VS_i^T	=	Volatile solids produced by cattle type T on a dry matter basis during the reporting period i (kg cattle ⁻¹ day ⁻¹)
C_i^T	=	Average number of animals for each cattle type T during the reporting period i
$PS_{AS,P,i}^T$	=	Percent of manure solids sent to (managed in) anaerobic manure storage/treatment system AS from cattle type T during the reporting period i . Dairy-specific data must be used. ¹⁹
t_i	=	Number of days in the reporting period i
0.8	=	System calibration factor ²⁰
$VS_{avail,AS,P,i-1}^T$	=	Volatile solids from cattle type T from the previous reporting period $i - 1$ available for degradation in anaerobic storage/treatment system AS (kg dry matter). ²¹
$VS_{deg,AS,P,i-1}^T$	=	Volatile solids from cattle type T that would have degraded during the previous reporting period $i - 1$ in anaerobic storage/treatment system AS in the absence of the project (kg dry matter).

$$VS_i^T = [DMI^T \times (1 - DE^T) + (UE^T \times DMI^T)] \times (1 - ASH^T) \quad (\text{Equation 5.29})$$

¹⁹ The value of PSAS,P,iT must take into account: the fraction of volatile solids that were recoverable (i.e., able to be collected for transfer into the anaerobic storage/treatment system), which is a function of the amount of time animals spent in different enclosure types; the type and efficiency the manure collection system; and the efficiency of the solid separation technology.

²⁰ From California Air Resources Board (2023) and US EPA (2016).

²¹ For the first reporting period of the project, if there are no site-specific data on the mass of volatile solids in the anaerobic storage/treatment system at the beginning of the project, the value may be calculated as described in Footnote 9. If the retention time for volatile solids in the anaerobic storage/treatment system is less than or equal to 30 days, set (VS_{avail,AS,P,i-1T} - VS_{deg,AS,P,i-1T}) to zero. Similarly, for any reporting period following the complete drainage and cleaning of solid buildup from the anaerobic storage/treatment system, (VS_{avail,AS,P,i-1T} - VS_{deg,AS,P,i-1T}) must be set to zero.

Where:

DMI^T	=	Dry matter intake of cattle type T in the reporting period (kg)
DE^T	=	Diet digestibility of the feed for cattle type T in the reporting period as a fraction of gross energy (Table 5.3) (%) Where DE is not available from primary data sources, projects may choose to use the Athian Dairy Total Digestible Nutrients Tool instead. Methodology for this tool is available as Appendix D. Athian Dairy Total Digestible Nutrients Tool Methodology .
UE^T	=	Urinary energy of cattle type T in the reporting period expressed as a fraction of gross energy (dimensionless). A default value of 4% is used for UE. This value can be reduced to 2% for cattle fed with 85 percent or more grain in their diet.
ASH^T	=	Ash content of feed for cattle type T in the reporting period (Table 5.3) (%)

DE and Ash are calculated as a weighted average using the data from Table 5.3. Project developers may provide alternative approaches to determine DE and Ash content of feed.

5.1.2.1.3 Project Manure Methane Emissions from Non-Anaerobic Storage/Treatment

$$CH_4^{T}_{dry,P} = \sum_{S,T} (C_i^T \times VS_i^T \times PS_{S,P}^T \times MCF_S \times B_0^T \times \rho_{CH_4} \times t) \quad (\text{Equation 5.30})$$

Where:

C_i^T	=	Average number of animals for each cattle type T during the reporting period
VS_i^T	=	Volatile solids produced by cattle type T on a dry matter basis during the reporting period (from (Equation 5.11, kg cattle ⁻¹ day ⁻¹))
$PS_{S,P}^T$	=	Percent of manure solids sent to (managed in) non-anaerobic manure storage/treatment system S from cattle type T during the reporting period. This should take into account the amount of manure that was recoverable, what fraction of recoverable manure was moved into the treatment/storage system, and what fraction of solids was separated. Dairy-specific data must be used. ²²
MCF_S	=	Methane conversion factor for non-anaerobic storage/treatment system S during the reporting period, based on T_2 , in Table 5.4(%)
B_0^T	=	Maximum methane-producing capacity of manure for cattle type T in Table 5.2 (m ³ CH ₄ kg VS ⁻¹)
ρ_{CH_4}	=	Density of methane = 0.67 (1 atm, 60°F) (kg m ⁻³)
t	=	Number of days in the reporting period

²² The value of PSS,PT must take into account the fraction of volatile solids deposited on land (e.g., pasture, open lot) and the fraction that was recoverable (i.e., able to be collected for transfer into a non-anaerobic storage/treatment system). These fractions are functions of the amount of time animals would have spent on pasture vs. different enclosure types. PSS,PT also depends on the type and efficiency of the manure collection system in collecting recoverable manure and the efficiency of the solid separation technology.

5.1.2.1.4 Project Manure Nitrous Oxide Emissions

Project emissions of N₂O are calculated as follows:

$$N_2O_{man,P}^T = \sum_T (N_2O_{dir,P}^T + N_2O_{ind,l,P}^T + N_2O_{ind,v,P}^T) \quad (\text{Equation 5.31})$$

Where:

$N_2O_{dir,P}^T$	=	Direct N ₂ O emissions from management of manure from cattle type T in the reporting period (kg N ₂ O)
$N_2O_{ind,l,P}^T$	=	Indirect N ₂ O emissions from leaching of manure from cattle type T in the reporting period (kg N ₂ O)
$N_2O_{ind,v,P}^T$	=	Indirect N ₂ O emissions from cattle type T from the volatilization of NH ₃ and NO _x in the reporting period (kg N ₂ O)

5.1.2.1.5 Project Direct N₂O Emissions from Manure Management

The calculation of direct N₂O emissions from the project entails multiplying the total amount of nitrogen excretion in each type of manure management system by an emission factor for that type of system and summing the obtained values. The emissions shall then be calculated using Equation 5.32:

$$N_2O_{dir,P}^T = (\sum_S (\sum_T (C^T \times N_{ex}^T \times AWMS_{S,TP}))) \times EF_{dir,S} \times \frac{44}{28} \quad (\text{Equation 5.32})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
N_{ex}^T	=	Average N excretion per animal of cattle type T (kg head ⁻¹)
$EF_{dir,S}$	=	Emission factor for direct N ₂ O emissions from non-anaerobic manure management system S (Table 5.5) (dimensionless)
$\frac{44}{28}$	=	Conversion factor between N ₂ O-N emissions and N ₂ O emissions
$AWMS_{S,TP}$	=	Project fraction of total annual volatile solids for each livestock category T that is managed in manure management system S.

5.1.2.1.6 Project Indirect N₂O Emissions from Leaching of Manure

The amount of nitrous oxide emitted from the project through leaching shall be calculated using Equation 5.33:

$$N_2O_{ind,l,P}^T = \sum_S (\sum_T (C^T \times N_{ex}^T \times AWMS_{S,TP} \times PL_S^T)) \times EF_l \times \frac{44}{28} \quad (\text{Equation 5.33})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
N_{ex}^T	=	Average N excretion per animal of cattle type T (kg head ⁻¹)
PL_S^T	=	Percent of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in non-anaerobic manure management system S (see Table 5.7) (%)

EF_l	=	Emission factor for N ₂ O emissions from nitrogen leaching and runoff. The default value is 0.011 kg N ₂ O-N (kg N leaching/runoff) ²³
$\frac{44}{28}$	=	Conversion factor between N ₂ O-N emissions and N ₂ O emissions
$AWMS_{S,TP}$	=	Project fraction of total annual volatile solids for each livestock category T that is managed in manure management system S.

5.1.2.1.7 Project Indirect N₂O Emissions from Volatilization of NH₃ and NO_x

The amount of nitrous oxide emitted from the project through volatilization in the forms of NH₃ and NO_x shall be calculated using Equation 5.34:

$$N_2O_{ind,v,P}^T = \sum_S (\sum_T (C^T \times N_{ex}^T \times AWMS_{S,TP} \times PV_S^T)) \times EF_v \times \frac{44}{28} \quad (\text{Equation 5.34})$$

Where:

C^T	=	Average number of animals for each cattle type T during the reporting period
N_{ex}^T	=	Average N excretion per animal of cattle type T (kg head ⁻¹)
PV_S^T	=	Percent of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x in non-anaerobic manure management system S (Table 5.7) (%)
EF_v	=	Emission factor for N ₂ O emissions from nitrogen volatilization. The default value is 0.010 kg N ₂ O-N (kg N leaching/runoff) ²⁴
$\frac{44}{28}$	=	Conversion factor between N ₂ O-N emissions and N ₂ O emissions
$AWMS_{S,TP}$	=	Project fraction of total annual volatile solids for each livestock category T that is managed in manure management system S.

5.1.2.2 Project GHG Emissions from Electricity Use

Project GHG emissions from electricity consumption are calculated by multiplying the total quantity of electricity consumed for manure management activities by the emissions factor for electricity for the state in which the dairy is located, using Equation 5.35. In the absence of manure management-specific data, total electricity consumption for the dairy may be used.

$$GHG_{elec,P}^T = Elec_P^T \times EF_{elec} \times \frac{1}{2.2046} \quad (\text{Equation 5.35})$$

Where:

$GHG_{elec,P}^T$	=	GHG emissions from electricity use for manure management activities for cattle type T during the reporting period (kg CO ₂)
$Elec_P^T$	=	Amount of electricity used for manure management activities for cattle type T during the reporting period (MWh)
EF_{elec}	=	Emissions factor for electricity in the state in which the dairy is located (lbs CO ₂ /MWh) ²⁵

²³ Additional emission factors may be used from Table 11.3 in Chapter 11 of IPCC (2019).

²⁴ Additional emission factors may be used from Table 11.3 in Chapter 11 of IPCC (2019).

²⁵ State-specific emissions factors are available via U.S. Energy Information Administration (2022). *US Electricity Profile 2021*. <https://www.eia.gov/electricity/state/>

2.2046 = Conversion factor from lbs to kg

5.1.2.3 Project GHG Emissions from Fuel Use

GHG emissions from fuel use in the reporting period are calculated by multiplying the total quantity of each type of fuel used and the emissions factor for that type of fuel, then summing the terms. If manure management-specific data are not available, total fuel use for the dairy may be used.

$$GHG_{fuel,P}^T = \sum_f (Fuel_{f,P}^T \times EF_{fuel,f}) \quad (\text{Equation 5.36})$$

Where:

$GHG_{fuel,P}^T$	=	GHG emissions from fuel use for manure management activities for cattle type T during the reporting period (kg CO ₂ eq)
$Fuel_{f,P}^T$	=	Quantity of fuel of each fuel type f consumed in the reporting period for manure management activities for cattle type T (MMBtu or gallon)
EF_f	=	Fuel-specific emission factor (Table 5.8) (kg/MMBtu or kg/gal)

5.1.3 AMMP Uncertainty

Quantitative uncertainty in both the model and variables has been assessed with a Monte Carlo simulation. This choice in technique was based on both Gold Standard and IPCC methodologies for assessing uncertainty and will be added in the form of a new addition to the final GHG reduction equation in the form of an uncertainty deduction. The uncertainties from the USDA's Quantifying Greenhouse Gas Fluxes Table 5.32 were used as the basis for the Monte Carlo simulation (See 0.) The method relies on known probability distributions for input variables and then uses those distributions to generate sets of sample input data to create a large volume of outputs that can then be modeled to show the distribution of the results against both the probability of that results occurrence and the frequency. The model itself identifies upper and lower bounds for each variable by each cattle type and using a simulated set of input data, identifies the degree of possible variation within those bounds and its effect on the final output.

The model created using the above methodologies was tested for 10,000 instances and the resultant uncertainty deduction that is to be applied to the results of Equation 4.2 is 56.21%.

5.2 Bovaer Quantification Methodology

5.2.1 Baseline GHG Emissions

Baseline GHG emissions from enteric fermentation follow the Intergovernmental Panel on Climate Change (IPCC, 2019) Tier 2 methodology. The baseline emissions are calculated according to the following equation:

$$GHG_{Bovaer,B}^T = CH_4 \times \frac{GWP}{1000} \quad (\text{Equation 6.1})$$

Where:

$GHG_{Bovaer,B}^T$	=	Baseline emissions from lactating dairy cattle during the reporting period (MT CO ₂ e)
CH_4	=	Baseline methane emissions for lactating dairy cattle during the reporting period (kg CH ₄)
GWP	=	Global warming potential of methane over a 100-year time frame; 27 is used per IPCC, 2021 (kg CO ₂ e per kg CH ₄)
1000	=	kg CO ₂ e per MT CO ₂ e

Baseline methane emissions for lactating dairy cattle during the reporting period are calculated according to the following equation:

$$CH_4 = \frac{(DMI \times GE) \times \left(\frac{Y_m}{100}\right) \times C \times t}{55.65} \quad (\text{Equation 6.2})$$

Where:

CH_4	=	Baseline enteric methane emissions for lactating dairy cattle during the reporting period (kg CH ₄)
DMI	=	Average daily dry matter intake for lactating dairy cattle during the reporting period (kg DM per head per day)
Y_m	=	Methane conversion factor for lactating dairy cattle during the reporting period (percentage of dietary gross energy lost as CH ₄ ; obtained from Table 5.1 adapted from Table 10.12 of IPCC, 2019)
55.65	=	Energy content of methane (MJ per kg CH ₄)
GE	=	Gross energy concentration of lactating dairy cattle diet (MJ per kg dry matter)
C	=	Average number of lactating dairy cattle during the reporting period (head)
t	=	Total number of days in the reporting period (days)

The appropriate Y_m is selected from Table 5.1 based on diet nutrient composition documented in UpLook using feed ingredient composition weighted averages and feed values from the NASEM (2021) feed library.

Table 6.1 Methane Conversion Factors (Y_m) adapted from Table 10.12 of IPCC, 2019

Lactating Dairy Cows	Y_m
DE $\geq$ 70 NDF $\leq$ 35	5.7
DE $\geq$ 70 NDF $>$ 35	6.0
DE 63-70 NDF $>$ 37	6.3
DE $\leq$ 62 NDF $>$ 38	6.5

DE= Diet digestible energy (%)

NDF= Diet neutral detergent fiber (%)

Please note Table 10.12 from IPCC, 2019 does not allow for appropriate Ym selection guidance if DE <70% and NDF ≤ 37%. Therefore, if DE% and NDF% constraints do not allow for appropriate Ym selection guidance with Table 5.1 (e.g., a diet with 69% DE and 36% NDF), a value of 5.85 is used.

5.2.2 Project GHG Emissions Quantification

Emissions in the project scenario are estimated as the sum of emissions during the reporting period from enteric fermentation and the production and transport of Bovaer according to the following equation:

$$GHG_{Bovaer,P}^T = [CH_4] \times \left[1 - \frac{|AF_H|}{100}\right] \times \left[\frac{GWP}{1000}\right] + GHG_{MT} \quad (\text{Equation 6.3})$$

Where:

$GHG_{Bovaer,P}^T$	=	Total project enteric CH ₄ emissions from lactating dairy cattle during the reporting period (MT CO ₂ e)
CH_4	=	Baseline enteric methane emissions for lactating dairy cattle during the reporting period (kg CH ₄)
AF_H	=	Herd enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer; Equation 5.4)
GWP	=	Global warming potential of methane over a 100-year time frame (kg CO ₂ e per kg CH ₄ ; IPCC, 2021)
1000	=	kg CO ₂ e per MT CO ₂ e
GHG_{MT}	=	Emissions associated with Bovaer manufacturing and transport during the reporting period (MT CO ₂ e)

5.2.3 Enteric Methane Adjustment Factor Quantification

The total lactating dairy cow herd enteric methane adjustment factor is determined by:

$$AF_H = AF \times PBCD \quad (\text{Equation 6.4})$$

Where:

AF_H	=	Herd enteric methane adjustment factor due to project implementation (% reduction in total lactating dairy cow herd methane due to feeding Bovaer)
AF	=	Enteric methane adjustment factor due to project implementation (% reduction in methane due to feeding Bovaer; Equation 5.6)
$PBCD$	=	Percent Bovaer-fed cow-days of total cow-days available (%; Equation 5.5)

Percent Bovaer-fed cow-days is calculated according to the following equation:

$$PBCD = \frac{\sum_i BC_i \times Bt_i}{C \times t} \quad (\text{Equation 6.5})$$

Where:

$PBCD$	=	Percent Bovaer-fed cow-days of total cow-days available (%)
BC_i	=	Average number of lactating dairy cattle in group i fed Bovaer during the reporting period (head)
Bt_i	=	Number of days in the reporting period Bovaer was fed to group i (days)
C	=	Average number of total lactating dairy cattle fed during the reporting period (head)
t	=	Total number of days in the reporting period (days)
i	=	Different groups of Bovaer-fed lactating dairy cattle where days and animal numbers differ between groups

A meta-analysis of 14 studies including 48 treatment means was utilized by Kebreab et al. (2023) to determine the effect of Bovaer on enteric methane in lactating dairy cattle. The effect of Bovaer on methane production is quantified in Equation 5.6 according to the Dose + NDF model in Table 2 of the meta-analysis which accounts for 3-NOP dose (mg/kg) and diet NDF (% DM). This equation was chosen in order to be consistent across baseline and intervention quantifications as the IPCC baseline equation does not account for dietary fat but does utilize diet NDF to determine the appropriate Ym tier.

Enteric methane adjustment factor due to intervention implementation is calculated according to the following equation:

$$AF = -32.8 - 0.285 \times (DOSE - 70.5) + 0.633 \times (NDF - 32.9) \quad (\text{Equation 6.6})$$

Where:

AF	=	Enteric methane adjustment factor due to intervention implementation (% reduction in methane due to feeding Bovaer)
-32.8	=	Overall mean estimate (intercept; % methane change) in Dose + NDF model in Table 2 of Kebreab et al. 2023
0.285	=	3-NOP dose factor in Dose + NDF model in Table 2 of Kebreab et al. 2023
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
70.5	=	3-NOP dose constant (mean dose; mg/kg) from Table 1 of Kebreab et al., 2023
0.633	=	NDF content factor from Dose + NDF model in Table 2 of Kebreab et al., 2023
NDF	=	Average diet neutral detergent fiber during the reporting period (% diet DM)
32.9	=	NDF content constant (mean content; % DM) from Table 1 of Kebreab et al., 2023

5.2.4 Product Manufacturing and Transport GHG Quantification

Emissions from the feed ingredient are estimated by including all GHG sources from manufacturing and transport according to the following equation:

$$GHG_{MT} = GHG_M + GHG_T \quad (\text{Equation 6.7})$$

Where:

GHG_{MT}	=	Emissions associated with Bovaer manufacturing and transport during the reporting period (MT CO ₂ e)
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GHG_M	=	Emissions associated with Bovaer manufacturing during the reporting period (MT CO ₂ e)
GHG_T	=	Emissions associated with Bovaer transport during the reporting period (MT CO ₂ e)

The total amount of product used during the reporting period is calculated according to the following equation:

$$Bovaer10 = \frac{DMI \times DOSE}{0.1 \times 1000000} \times PBCD \times C \times t \quad (\text{Equation 6.8})$$

Where:

$Bovaer10$	=	Amount of Bovaer10 used during the reporting period (kg)
DMI	=	Average daily dry matter intake during the reporting period (kg DM per head per day)
$DOSE$	=	Dose of 3-NOP fed during the reporting period (mg 3-NOP per kg diet DM)
$PBCD$	=	Percent Bovaer-fed cow-days of total cow-days available (%)
C	=	Average number of lactating dairy cattle fed Bovaer during the reporting period (head)
t	=	Number of days in the reporting period (days)
0.1	=	Active ingredient content of Bovaer10 (kg 3-NOP per kg Bovaer10)
1000000	=	mg 3-NOP per kg 3-NOP

Manufacturing-based GHGs associated with product (Bovaer) production are supplied by the manufacturer based on ISO standards 14040.44:2006 and ISO 14067:2018. The deduction to intervention GHG reductions (Equation 5.9) is calculated according to the following equation:

$$GHG_M = \frac{Bovaer10 \times PCF}{1000} \quad (\text{Equation 6.9})$$

Where:

GHG_M	=	Manufacturing-based GHG emissions due to Bovaer use during the reporting period (MT CO ₂ e)
$Bovaer10$	=	Amount of Bovaer10 used during the reporting period according to Equation 5.8 (kg)
PCF	=	Product Carbon Footprint of Bovaer10 production (4.84 kg CO ₂ e per kg Bovaer10)
1000	=	kg CO ₂ e per MT CO ₂ e

Emissions from the transport of the feed ingredient to the project site are calculated as:

$$GHG_T = \frac{Bovaer10 \times (EFT_R + EFT_O)}{1000} \quad (\text{Equation 6.10})$$

Where:

GHG_T	=	Emissions associated with Bovaer transport during the reporting period (MT CO ₂ e)
$Bovaer10$	=	Amount of Bovaer10 used during the reporting period according to Equation 5.8 (kg)

EFT_R	=	Emissions factor from road transportation of Bovaer10 during the reporting period (0.6097 kg CO ₂ e per kg Bovaer10)
EFT_O		Emissions factor from ocean transportation of Bovaer10 during the reporting period (0.2702 kg CO ₂ e per kg Bovaer10)
1000	=	kg CO ₂ e per MT CO ₂ e

Road transportation of Bovaer is estimated using Environmental Protection Agency 2024 GHG Emissions Factors Hub data (Scope 3 Category 4: Upstream Transportation and Distribution and Category 9: Downstream Transportation and Distribution) including CO₂, CH₄, and N₂O from medium- and heavy-duty truck emissions. Ocean transport is estimated using Department for Environment, Food and Rural Affairs, Annex 7 Freight Transport Conversion Tables and includes refrigerated cargo with the highest GHG emissions in order to be conservative.

Distances of travel are considered the same for all farms implementing Bovaer and are thus conservatively estimated as the distance for product to be delivered to Tulare, CA. This location was chosen because it contains the largest population of cows the furthest distance from the Elanco warehouse in Clinton, IN.

5.2.5 Bovaer Uncertainty

The major risk that could substantially affect the intervention's GHG emission reductions is in the estimation of feed intake. Traditionally a large amount of uncertainty in dairy operation greenhouse gas calculations originates from estimations of feed intake and digestibility, which vary widely between and within cattle and geographies. A significant portion of this uncertainty is mitigated in this program through utilizing each operation's individual feed ingredient data to quantify dry matter intake and use contemporaneous estimates of gross and digestible energy for the quantification of both baseline and intervention emissions. The availability of these data facilitates a precise measure of gross energy intake from which enteric emissions reductions are calculated. Because feed intake is recorded and received directly from the dairy farm, the remaining uncertainty associated with the GHG reductions quantified in this program arise from the adjustments made around the intervention of Bovaer implementation. These adjustments are taken directly from a peer-reviewed meta-analysis. Bovaer has reliable and substantiated data regarding its mode of action and effect.

Because uncertainty associated with feed intake, a primary driver in GHG emissions, was mitigated through incorporating operation-specific feed data in a tier 2 approach to calculate GHG emissions and uncertainty associated with adjustment factors for Bovaer's effect were mitigated using a meta-analysis, no deductions for uncertainty will be taken. Outside of the product carbon footprint and transportation emissions (PCF, EFTR, and EFTO), which have minimal impact to the overall reductions, the adjustment factor is the major parameter Bovaer affects within emissions calculations. All other variables are the same for both Baseline and the Project.

6 Deviations from Protocol Methodologies

Deviations from the methodologies in Sections 5 of this protocol are not allowed.

7 Monitoring

This program is 100% monitoring. All producers participating in the program will go through verification of their baseline data and verification of monitoring periods. A monitoring plan has been developed for all monitoring and reporting activities associated with the project, standardized across all participating farms.

Verifiers will use the monitoring plan and report to confirm that the requirements of this program have been met. This monitoring plan provides the processes, requirements, and sources of information necessary to assess the GHG reductions created by the practices included in this protocol.

This includes:

1. The procedures for collecting data on intervention activities related to implementation.
2. The data points collected to verify emission reduction, project, and baseline calculations.
3. The QC/QA processes to ensure the accuracy and consistency of the data collected.

The monitoring reports described in the monitoring plan include the following elements:

1. General description of the project, including the location of the cattle operations
2. List of the practices implemented
3. Description of the process and frequency of data collection and the archiving procedures
4. Recordkeeping plan
5. Role of any individuals performing activities related to the practices implemented
6. Quality assurance/quality control (QA/QC) procedures to ensure the accurate collection and entry of data in quantification systems
7. Monitoring reports must include the monitoring time period.
8. Monitoring reports must include the list of parameters measured and monitored.
9. Monitoring reports must include the types of data and information reported, including units of measurement.
10. Monitoring reports must include the origin of the data.
11. The monitoring report must include an attestation as to regulatory compliance.
12. The monitoring report should be submitted no less frequently than annually and no more frequently than 30 days.
13. The monitoring period can be as short as 30 days. The maximum monitoring period is 12 months.
14. The monitoring report must be submitted and shared with Athian, as the program administrator.

A monitoring period represents a calendar month. Where necessary, multiple months may be combined at the discretion of the verifier. When the monitoring period is over (i.e. the month

has fully passed), data from operating records are input into the Athian quantification tool to assess the impact of the intervention for that monitoring period. Supporting documents are collected concurrently. Once all required inputs and supporting documentation are collected, a third-party verifier receives the information to assess the validity of the reported inputs as well as verify the quantification themselves. Participating producers are given the opportunity to produce further documentation of any values the verifier determines to be non-compliant before a final decision on the results of the monitoring period is rendered, either verified or not.

This monitoring plan provides the requirements and sources of information necessary to assess the GHG reductions created using this protocol. This monitoring plan describes the procedures for collecting data on intervention activities related to the implementation of protocol practices. The data collected will support emission reduction and baseline calculations. This monitoring plan also outlines the QC/QA processes to ensure the accuracy and consistency of the data collected that will be used to verify emissions reduction outcomes.

7.1 Data Quality Assurance

AMMP and Bovaer share three common data elements. Those are DMI, headcount, and days in the reporting period. In order to address consistency between the two programs, these three elements will need to be consistent across both when the program is being verified, and as such will need to use the same source. Because Bovaer requires Uplook enrollment, Uplook will be the source of truth for this information in concurrent enrollment situations. DMI and headcount for non-Bovaer enrolled animals (i.e. dry and heifers) may come from Uplook but may also come from an alternative source. The same standard of verification will be applied to both self-reported information and to Uplook data, as is current Athian and verifier practice.

The Athian Data Quality Management Plan aims to ensure that a producer's data is accurate, reliable, and fit for its intended purpose to assess the impact of the practices included in this protocol on CO₂e emissions associated with the management of manure and feed cultivation. The goals and objectives of a can be categorized into several key areas, each targeting different aspects of data quality management. These include accuracy, timeliness, comparability, and creditability.

Accuracy:

- **Data Collection Methods:** Data will be provided by the farm directly based on various on farm systems.
- **Consistency Checks:** Input forms will check for data type and range preventing grossly invalid data from being entered.
- **Method Validation:** Based on type, input may be limited to certain ranges or values. Additionally, producers must attest to and confirm accuracy.

Timeliness:

- **Data Collection Frequency:** As defined by the protocol
- **Data Reporting Schedule:** Specific schedules are defined by the protocol monitoring plan.

- Response Procedures for Data Variations: Significant data issues should be prevented at entry. Additionally identified issues can be corrected by Athian staff as needed.

Comparability:

- Standardized Methods Used: Form input is used to collect quantitative data from on farm systems.
- Benchmarking: Per the Athian Data Retention Policy, all GHG related data is kept for a minimum of 10 years. Data, in aggregated and anonymized form, can be used for benchmarking if/when applicable.

Creditability:

- Documentation of Data Processes: All data processes are ultimately governed by the Athian Data Protection, Data Retention, and Software Development Lifecycle (SDLC) policies. These policies are maintained as controlled documents in the Athian compliance system (Draata) and are reviewed and updated at least annually.
- Transparency Measures: Data transparency is critical to credibility and integral to the data collection process. Data input directly in the platform from producers requires an attestation from the producer as to the accuracy before being submitted for verification. Data collected via an integration to a 3rd party data collection software also requires the producer to attest to the accuracy. In addition, producers have visibility as to the data provided to 3rd party verifiers and can see the status of the verification of each element of data submitted. All verification reports include each data element collected and reviewed as part of the verification process for complete transparency in the reporting of the emissions result.

The Athian platform has a comprehensive set of automated processes that confirm the integrity, correctness, and completeness of data. These include checking the data upon ingestion from any 3rd party data source, inclusive of data delivered via API or manually entered, for completeness and accuracy. These checks include verification of appropriate formatting, field-level requirements that ensure the presence of all required data, and identification of any data variance from the previously verified data. If errors are identified, notifications are generated and delivered to engineering, product management, and service management for resolution. Those parties then determine the source and scope of the issue(s), engage any necessary participating party, resolve and document the identified issues.

In addition to the data validation checks identified above, Athian has implemented a service-driven approach for applying logic consistently, significantly reducing the potential for error in the process. The programmatic logic used reduces or replaces much of the process that is prone to human error. The Athian platform hosts the mechanisms for documenting any data discrepancy as well as their respective severity and solution. The platform retains a complete transaction history for all data ingested, inclusive of date/time stamp and the individual user or software supplying the information. This ensures that Athian will have a complete history/picture of all data used when rendering a decision or result.

All data used to meet GHG carbon accounting standards for impact units must be retained for a minimum of ten years. This includes producer business contact information, location information, monitoring period information, and all verification information. All of the aforementioned processes and procedures adhere to industry best practices, including SOC 2 review. The quantification tool for this program is thoroughly tested against known results of data sets any time updates to the quantification methodology or tool are made. The tests

follow the same methods as used for the Simulated Quantification Model and are checked against the quantification methodology for accuracy by the Athian development team.

Table 6.1 Monitoring parameters

Data/ Parameter	Description	Data Unit	Data Source	Measurement Frequency	Values Applied	Measurement methods and procedures, including QA/QC	Roles and Responsible	Data Management
T	Type of cattle included in the project	Cattle type	Operating records	Every reporting period	Type of cattle	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
GWP _{CH4}	Global warming potential of methane	kg CO ₂ e kgCH ₄₋₁	Reference	N/A	29.8	Constant	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool	Stored within the Athian platform and available as part of the published protocol.
GWP _{N2O}	Global warming potential of nitrous oxide	kg CO ₂ e kg N ₂ O ⁻¹	Reference	N/A	273	Constant	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool	Stored within the Athian platform and available as part of the published protocol.
AS	Anaerobic storage/treatment system	Type of treatment system	Operating records	At project start, and confirmed every reporting period	Name of system(s)	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
B ₀ ^T	Maximum methane-producing capacity of manure for cattle type T	m ³ CH ₄ kg VS ⁻¹	Reference	N/A	Table 5.2	Constant	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored and available as part of the published protocol.

f_i	The van't Hoff-Arrhenius factor for the current reporting period i	kg m^{-3}	Calculated	Every reporting period	Calculation based on temperature information for the current reporting period	N/A	Athian is responsible for ensuring this value is correctly calculated in their platform with regular checks of the quantification tool.	Data located in the Athian platform and available to producers and third-party verifiers.
T_2	Average ambient temperature in the reporting period	K	Obtained via API with the National Weather Service based on producer location	Every reporting period	Approx. 222 to 323 K	Obtained from the closest weather station within the air basin as reported by major weather outlets such as the National Weather Service.	Athian is responsible for ensuring this value is correctly brought into their platform with regular checks of the quantification tool.	The Athian platform is intended to interface with the National Weather Service to pull this information in its most accurate form for each monitoring period.
VS_i^T	Volatile solids produced by cattle type T on a dry matter basis during the reporting period i	$\text{kg cattle}^{-1} \text{ day}^{-1}$	Calculated	Every reporting period	Approx. 5.4 to 7.7	N/A	Athian is responsible for ensuring this value is correctly calculated in their platform with regular checks of the quantification tool.	Data located in the Athian platform and available to producers and third-party verifiers.
C_i^T	Average number of animals for each cattle type T during the reporting period i	Number of cattle	Operating records	Every reporting period	Number of cattle	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.

PS _{AS,B,T}	Percent of manure solids that would have been sent to (managed in) anaerobic manure storage/treatment system AS from cattle type T during the reporting period i in the absence of the project.	%	Operating records	Every reporting period	0 to 100%	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
t _i	Number of days in the reporting period i	days	Measured	Every reporting period	≥28	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
DMI _T	Dry matter intake (DMI) of cattle type T	kg	Operating records	Every reporting period	Approx. 19 to 23	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
DE _T	Diet digestibility of cattle type T in the reporting period as a fraction of gross energy	%	Operating records or reference	Every reporting period	Table 5.3	Obtained from producer records If not available from operating records, then based on the information in Table 5.3	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.

UE _T	Urinary energy of cattle type T in the reporting period expressed as a fraction of gross energy	dimensionless	Reference	Every reporting period	A default value of 4% is used for UE. This value can be reduced to 2% for cattle fed with 85 percent or more grain in their diet.	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
ASH _T	Ash content of feed for cattle type T in the reporting period	%	Reference	Every reporting period	Table 5.3	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
S	Non-anaerobic manure storage/ treatment system	Type of treatment system	Operating records	Every reporting period	Name of treatment system. Listed in tables 5.4 and 5.5	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
VS _T	Volatile solids produced by cattle type T on a dry matter basis during the reporting period	kg cattle ⁻¹ day ⁻¹	Calculated	Every reporting period	Approx. 5.4 to 7.7	N/A	Athian is responsible for ensuring this value is correctly calculated in their platform with regular checks of the quantification tool.	Data located in the Athian platform and available to producers and third-party verifiers.

PS _{S,BT}	Percent of manure solids that would have been sent to (managed in) non-anaerobic manure storage/treatment system S from cattle type T during the reporting period in the absence of the project.	%	Operating records	Every reporting period	0 to 100%	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
MCF _S	Methane conversion factor for storage/treatment system S	%	Reference	Every reporting period	Table 5.4	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
EF _{dir,S}	Emission factor for direct N ₂ O emissions from non-anaerobic manure management system S	dimensionless	Reference	Every reporting period	Table 5.5	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
CP _T	Crude protein content in the overall diet	%	Operating records	Every reporting period	0 to 100%	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.

M	Average milk production per cow during the reporting period	kg head ⁻¹ day ⁻¹	Operating records	Every reporting period	25 to 31	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
M _p	Percentage of protein in the milk during the reporting period	%	Operating records	Every reporting period	0-100%	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
WG _T	Average weight gain of the animals in the population of cattle type T	kg day ⁻¹	Reference	Every reporting period	Table 5.6-C	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
BW _T	Average live body weight (BW) of the animals in the population of cattle type T	kg	Reference	Every reporting period	Table 5.6-B	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
MW _T	Mature live body weight of an adult animal in moderate body condition	kg	Reference	Every reporting period	Table 5.6-A	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.

PL _{ST}	Percent of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in non-anaerobic manure management system S	%	Reference	Every reporting period	Table 5.7 0 to 6%	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
EF _(l)	Emission factor for N ₂ O emissions from nitrogen leaching and runoff.	kg N leaching/ runoff	Reference	Every reporting period	The default value is 0.011 kg N ₂ O-N.	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
EF _(v)	Emission factor for N ₂ O emissions from nitrogen volatilization.	kg N leaching/ volatilization	Reference	Every reporting period	The default value is 0.010 kg N ₂ O-N.	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
PV _{ST}	Percent of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x from non-anaerobic manure management system S	%	Reference	Every reporting period	Table 5.7	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.

Elec _{BT}	The average amount of electricity that would have been used for manure management activities for cattle type T in the absence of the project, based on average electricity consumption for the same time as the reporting period from the 2 years prior to the project start date.	MWh	Operating records	Every reporting period		Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
EF _{elec}	Emissions factor for electricity in the state in which the dairy is located	lbs CO ₂ MWh ⁻¹	Reference	Every reporting period	U.S. Energy Information Administration (2022). <i>US Electricity Profile 2021</i> . https://www.eia.gov/electricity/state/	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.

Fuel _{BT}	The average amount of fuel that would have been used for manure management activities for cattle type T in the absence of the project, based on average fuel consumption for the same time as the reporting period from the 2 years prior to the project start date	mmBtu or gallon	Operating records	Every reporting period		Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
EF _{fuel, f}	Fuel-specific emission factor	kg/mmBtu or kg/gal	Reference	Every reporting period	Table 5.8	N/A	Athian is responsible for ensuring this value is correct in their platform with regular checks of the quantification tool.	Stored within the Athian platform and available as part of the published protocol.
PS _{AS,P,iT}	Percent of manure solids sent to (managed in) anaerobic manure storage/treatment system AS from cattle type T during the reporting period i.	%	Operating records	Every reporting period	0 to 100%	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.

PS _{S,PT}	Percent of manure solids sent to (managed in) non-anaerobic manure storage/treatment system S from cattle type T during the reporting period.	%	Operating records	Every reporting period	0 to 100%	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
Elec _{PT}	Amount of electricity used for manure management activities for cattle type T during the reporting period	MWh	Operating records	Every reporting period		Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
Fuel _{PT}	The average amount of fuel used for manure management activities for cattle type T in the reporting period	mmBtu or gallon	Operating records	Every reporting period		Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
MF	Percentage of fat in the milk during the reporting period	%	Operating records	Every reporting period	3.5 to 4.5	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.

AWMS _{S,TB}	Baseline fraction of total annual volatile solids for each livestock category T that is managed in manure management system S.	%	Operating records	Every reporting period	0 to 100	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
AWMS _{S,Tp}	Project fraction of total annual volatile solids for each livestock category T that is managed in manure management system S.	%	Operating records	Every reporting period	0 to 100	Obtained from producer records	Producer: inputting these values and ensuring they are correct Verifier: corroborating values against documentation	Data located in the Athian platform and available to producers and third-party verifiers.
PBCD	Percent Bovaer-fed cow-days	Cow-days	Primary data obtained from farm operating records	Every reporting period	Calculated as total Bovaer-fed cow-days divided by total cow-days available in reporting period (month). Total Bovaer-fed cow days accounts for the number of cows fed Bovaer and the number of days they were fed Bovaer during the reporting period.	Values are audited at verification	Producer/Farm Operator: Responsible for feeding Bovaer and recording the number of cows and days it is fed during the reporting period and calculating PBCD and entering it in UpLook. Elanco: Responsible for storing farm data via UpLook, calculating total Bovaer use during reporting period and reporting data to Athian	Data stored in UpLook

DE	Digestible energy; Diet digestible energy as a % of diet gross energy in the lactating cattle diet	% of gross energy	Calculated value using primary data from farm records & secondary data on chemical composition of feedstuffs	Every reporting period	Diet digestible energy composition is documented in UpLook using feed ingredient composition weighted averages and digestible energy values from the NASEM (2021) feed library	Calculated value in UpLook that is compared to an expected range (i.e., min and max acceptable values). Because the NASEM values used are standardized, the only way an alert should be triggered is if the user entered an unrealistic ingredient mix.	Elanco: Responsible for calculating DE value of diet, verifying that it is an acceptable value, and storing farm data via UpLook	Data stored in UpLook
NDF	Neutral detergent fiber; a measure of the fiber concentration of the lactating dairy cattle diet	% of diet dry matter	Calculated value using primary data from farm records & secondary data on chemical composition of feedstuffs	Every reporting period	Diet neutral detergent composition is documented in UpLook using feed ingredient composition weighted averages and neutral detergent fiber values from the NASEM (2021) feed library	Calculated value in UpLook that is compared to an expected range (i.e., min and max acceptable values). Because the NASEM values used are standardized, the only way an alert should be triggered is if the user entered an unrealistic ingredient mix.	Elanco: Responsible for calculating NDF value of diet, verifying that it is an acceptable value, and storing farm data via UpLook	Data stored in UpLook

GE	Gross energy concentration of the lactating dairy cattle diet	MJ per kg diet dry matter	Calculated value using primary data from farm records, secondary data on chemical composition of feedstuffs, and NASEM Equations	Every reporting period	Diet gross energy composition is documented in UpLook using feed ingredient composition weighted averages, feed chemical compositions from the NASEM (2021) feed library, and Equations 3-1 and 3-2 from NASEM (2021)	Calculated value in UpLook that is compared to an expected range (i.e., min and max acceptable values). Because the NASEM values used are standardized, the only way an alert should be triggered is if the user entered an unrealistic ingredient mix.	Elanco: Responsible for calculating GE value of diet, verifying that it is an acceptable value, and storing farm data via UpLook	Data stored in UpLook
Ym	Percentage of gross energy converted to methane	Methane conversion factor	Secondary data from IPCC, 2019 and adapted to Table 5.1	Every reporting period	The appropriate Ym is selected from Table 5.1 based on diet digestible energy (DE) and neutral detergent fiber (NDF) of the diet	Values are audited at verification	Elanco: Responsible for selecting correct Ym based on calculated DE and NDF values and storing farm data via UpLook.	Data stored in UpLook
GWP	Global warming potential of methane over 100-year time horizon	kg CO _{2e} per kg CH ₄	Secondary data from Sixth Assessment Report from IPCC, 2021 (AR6)	Annually	27	Values are audited at verification	Elanco: Responsible for updating value in UpLook when deemed necessary	Data stored in UpLook

AF	Enteric methane adjustment factor due to Bovaer implementation	% reduction in methane due to feeding Bovaer	Calculated value using dose (primary farm data) and NDF (calculated parameter)	Every reporting period	Calculated value using Dose + NDF model in Table 2 from the Kebreab et al., 2023 meta-analysis of 3-NOP effects on enteric methane in lactating dairy cows and incorporating dose (primary farm data) and NDF (calculated value) for each reporting period	Values are audited at verification	Producer/Farm Operator: Responsible for feeding of Bovaer according to the label, data recording and entry in UpLook Elanco: Responsible for calculating value and storing farm data via UpLook	Data stored in UpLook
DOSE	Dose of 3-nitrooxypropano I (3-NOP) fed; Bovaer10 is 10% 3-NOP	mg 3-NOP per kg diet dry matter	Primary data obtained from farm records	Every reporting period	Any value allowed, but if value is not on-label (60-80 mg/kg DM), then AF is set equal to zero.	Values are audited at verification	Producer/Farm Operator: Responsible for feeding of Bovaer according to the label, providing records of proof of the dose fed, and data recording and entry in UpLook Elanco: Responsible for calculating the impact of Bovaer fed, storing farm data via UpLook, and producing and distributing Bovaer	Data stored in UpLook

PCF	Carbon footprint associated with manufacturing Bovaer10	kg CO ₂ e per kg Bovaer10	Secondary data originating from manufacturer	Annually	4.84	Values are audited at verification	Elanco: responsible for reviewing/updating product carbon footprint at annual protocol review	Data stored in UpLook
EFT _R	Carbon footprint associated with land transportation of Bovaer10	kg CO ₂ e per metric-ton mile for Bovaer10	Calculated data originating from secondary sources	Annually	0.18668	Values are audited at verification	Elanco: responsible for reviewing/updating product transportation carbon footprint at annual protocol review.	Data stored in UpLook
EFT _O	Carbon footprint associated with ocean transportation of Bovaer10	kg CO ₂ e per metric-ton mile for Bovaer10	Calculated data originating from secondary sources	Annually	0.06994	Values are audited at verification	Elanco: responsible for reviewing/updating product transportation carbon footprint at annual protocol review.	Data stored in UpLook

8 Reporting

Project developers must provide the following documentation each reporting period to generate credits from this protocol:

1. Name and address of the project developer
2. List of all of the operations included in the project including the owner/operator contact information and address of the operation
3. Regulatory compliance documentation and attestation
4. Monitoring plan
5. Monitoring report with all the data used in the calculations for Section 5 of the protocol
6. Monitoring report must include the intended use and user of the monitoring report.

8.1 Record Keeping

For purposes of third-party verification and historical documentation, project developers must keep all information listed in this protocol for a period of 10 years after the information is generated. The information the project developer should retain includes:

1. All data inputs for the calculation of the project emission reductions as well as the results of emission reduction calculations
2. Copies of all permits, Notices of Violations (NOVs), and any relevant administrative or legal orders dating back at least 3 years prior to the project start date
3. All verification records and results
4. All maintenance records relevant to the monitoring equipment

9 Verification

Verification bodies will contract directly with Athian for all validation and verification engagements.

Projects verified under this protocol will meet, at minimum, the auditing standard of limited assurance and adhere to 14064-3. The verification body must provide a factual statement expressing the outcome of the verification.

Issues identified during verification must be classified by verification bodies as either material (significant) or immaterial (insignificant). To be verified successfully, all reported emissions reductions must be free of material misstatements.

All projects developed under this protocol must achieve >95 percent level of accuracy. This means that the project's calculated emission reductions must be less than 5 percent different than those calculated by the verifier.

9.1 Verification Body Requirements

To conduct verification under this protocol, all Validation and Verification Bodies (VVB) must meet the following criteria:

1. Accreditation under International Organization for Standardization (ISO) 14065: 2013 with conformance to all accreditation requirements under ISO 14065, ISO 14064-3: 2006, IAF MD 6: 2014 and all other accreditation requirements, or Acceptance in the American National Standards Institute (ANSI) accreditation program, having filed a full application for ISO 14065: 2020
2. Demonstrated/documented subject matter expertise in the on-farm operations related to an approved protocol (e.g., Dairy Operations; Feed Lot Operations)
3. Demonstrated/documented experience in a particular region or state where the verification will occur
4. Monitoring conducted in accordance with the requirements of the relevant protocol
5. Monitoring conducted in a manner that allows for a complete and transparent quantification of GHG reductions

9.2 Conflict of Interest

When conducting verification under this protocol verifiers must be seen as credible, independent, and transparent. To meet this requirement, a conflict of interest (COI) determination must be made prior to starting any verification activities. A COI occurs in any situation that compromises the verifier's ability to perform an independent verification. Every verifier must provide information about its organizational relationships, internal structures, and management systems for identifying potential COIs. Verifiers must evaluate any potential conflicting services it has provided to the project developer, including any advice or consulting provided outside of the verification process.

9.3 Verification Process

To verify the project, the verifier must develop a risk-based verification plan that considers the size and complexity of the project and the relevant sector, technology, and processes. The verifier must follow the following process:

1. Complete a COI evaluation. If there is a potential COI, the verifier is not allowed to conduct the verification.
2. Prepare a verification plan that includes, at a minimum:
 - a. A list of people from the VVB involved in the verification
 - b. A list of the location and dates of any on-site visits that will be conducted
 - c. The types of data and documents that will be reviewed by the verifier
 - d. A list of the people who are expected to be interviewed as a part of the verification
3. Conduct a kick-off meeting with all parties to lay out the timeline and process of the verification.
4. Conduct, at minimum, one annual on-site visit to confirm practice implementation

5. Undertake a desk review of the data from the project.
6. Prepare a verification report that includes:
 - a. A verification statement documenting the outcome of the assessment (reduction results) and if there was any material discrepancy noted
 - b. Key details about the project including: producer and farm operation identification, verifying body and lead verifier contact information, protocol information, and intervention information
 - c. A description of the protocol, the objectives and criteria used to arrive at the final result, the scope of the project, the level of assurance associated with the project, and any details about the implementation of the practices observed
 - d. Detail about the verification process used to complete the assessment including approach and methods and also noting any conflict of interest
 - e. Verification findings including confirmation of producer eligibility, adherence to the criteria established in the protocol, the verified emissions quantification values, and the final written opinion of the verifier(s)
 - f. An issue log capturing any issues identified during the verification and their classification as either material (significant) or immaterial (insignificant)
 - g. A representation of all data / documents used in the process of verification

10 Practice Definitions

Closed Solar Drying: Unlike open solar drying described above, closed solar drying involves putting manure into an enclosed container for its drying process. This improves the efficiency of drying and reduces the risk of leaching and runoff. In both practices, manure emissions are reduced through the drying of manure volatile solids.

Compost Bedded Pack Barn: Bedded pack barns are a loose housing system in which dairy cattle have an open area in which to rest, ruminate, and spend time, usually separated from the area where they eat and drink. The housing area (the pack) may be managed either aerobically or anaerobically. Anaerobic management includes the regular re-application of fresh bedding materials such as straw or sawdust without any aeration or turning. Aerobic or compost barns are stirred regularly (often multiple times a day) which helps with moisture and temperature control in the compost. Manure is deposited directly into the bedding material, which serves to distribute moisture evenly away from the manure solids and acts as additional dry matter for the compost build up in the pack system itself. Previously composted manure may also be used as the pack bedding material, often in conjunction with straw or sawdust to improve mixing.

Composting in Vessel: Composting is the decomposition of organic material in a managed system. Composting in a vessel controls the environment of the composting by containing it to a manageable space. The success of this practice in emissions reductions hinges on the forced aeration and/or mixing. These techniques not only help control the moisture content and microorganism load of the compost, but also regularly introduce oxygen to ensure a non-anaerobic environment for decomposition.

Composting in Windrows: As with composting in vessels, composting in windrows is the decomposition of organic material in a managed system. This particular style of composting builds up lines of compostable material for this degradation. Exposure to the elements and the sun helps ensure a non-anaerobic environment for this degradation, however additional mixing and aeration may be used as well.

Conversion From A Flush To Scrape Manure Collection System: A flush system involves the use of water to regularly flood concrete lanes in a barn ("flushing" them) to remove manure, urine, and other debris that may have accumulated. This flush then travels into a containment area, a lagoon, for breakdown. The addition of water to this system ensures an anaerobic environment for the manure solids once they do reach that containment area unless the solids are first separated out. A scrape system, in contrast, uses equipment such as tractors or an automated version attached to a tether and motor to remove manure from the lanes. These systems introduce less overall moisture to the manure solids and when used in conjunction with a solid separation practice from the list below, result in a reduced amount of manure solids that are managed anaerobically.

Daily Spread: Daily spread involves the removal of manure from the confinement facility and direct land application within 24 hours. This practice diverts manure solids from anaerobic conditions in lagoons, but may also have some of the benefits of solar drying in further separating the liquids from the manure solids.

Forced Evaporation With Natural Gas Fueled Dryers: Much like solar drying, this practice relies on the separation of manure solids and liquids to reduce the amount of solids managed anaerobically. These dryers force heated air into contact with the manure, drying it out more quickly than in the solar drying options.

Open Solar Drying: This practice relies on the use of passive solar energy to remove moisture from the manure. Manure is spread out in a layer over a space where it can get sufficient sun exposure to dry it out before being moved elsewhere for either composting or land application. In both solar drying practices, manure emissions are reduced through the drying of manure volatile solids. This practice also applies to dry lot housing.

Pasture-Based Management: Managing dairy animals on pasture instead of in confinement facilities allows for emissions reductions in several ways. Manure is applied to the land directly immediately upon excretion, diverting it from anaerobic degradation. On pasture, the manure is spread out, aerating it, by the movement of animals as well as environmental factors. This exposes the manure further to oxygen. Additionally, sun exposure increases the rate of drying on land, further separating the manure solids from the liquids, ensuring non-anaerobic degradation.

Slatted Floor Pit Storage: In a slatted floor pit storage facility, cattle are housed over a floor with openings into a storage container beneath the barn itself. The manure is stored in this space for 30 days or less, after which it is cleaned out. While this storage method does not involve separation of liquids and volatile solids, which does create an anaerobic environment, the regular cleaning greatly reduces the build-up of volatile solids that might completely degrade in this anaerobic environment as opposed to a large lagoon. After the cleaning takes place, the manure may be relocated for compost or for direct land application.

Solid Separation Of Manure Solids Before Being Deposited Into An Anaerobic Environment: Solid separation technologies include but are not limited to weeping walls, stationary screens, vibrating screens, screw presses, centrifuges, roller drums, belt presses/screens, advanced solid-liquid separation assisted by flocculants and/or bead filters, and vermifiltration.

Solid Separation With An Aerated Vermifiltration System: Vermifiltration uses a filter system of several layers, the topmost of which is earthworms in a combination of soil and vermicompost. This layer aids in the separation of manure solids and liquids in the same way as the vermicomposting does, and the resultant by-products may be used in compost and agricultural applications. This method, like the other solid separation methods, diverts manure solids from anaerobic degradation into an aerobic system, where the worms assist in the aeration of the manure while in the filter.

Solid Storage: This practice involves the storage of manure in unconfined piles or stacks, diverting manure from liquid storage in a lagoon. These stacks must be cleaned out with some regularity to ensure the minimization of anaerobic storage. From this storage, manure may be added to compost or spread on cropland or pasture.

11 References

Athian Alternative Manure Management Protocol, December 2025

Athian Bovaer Protocol, September 2024

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Appendix A. Table of AMMP Variable References

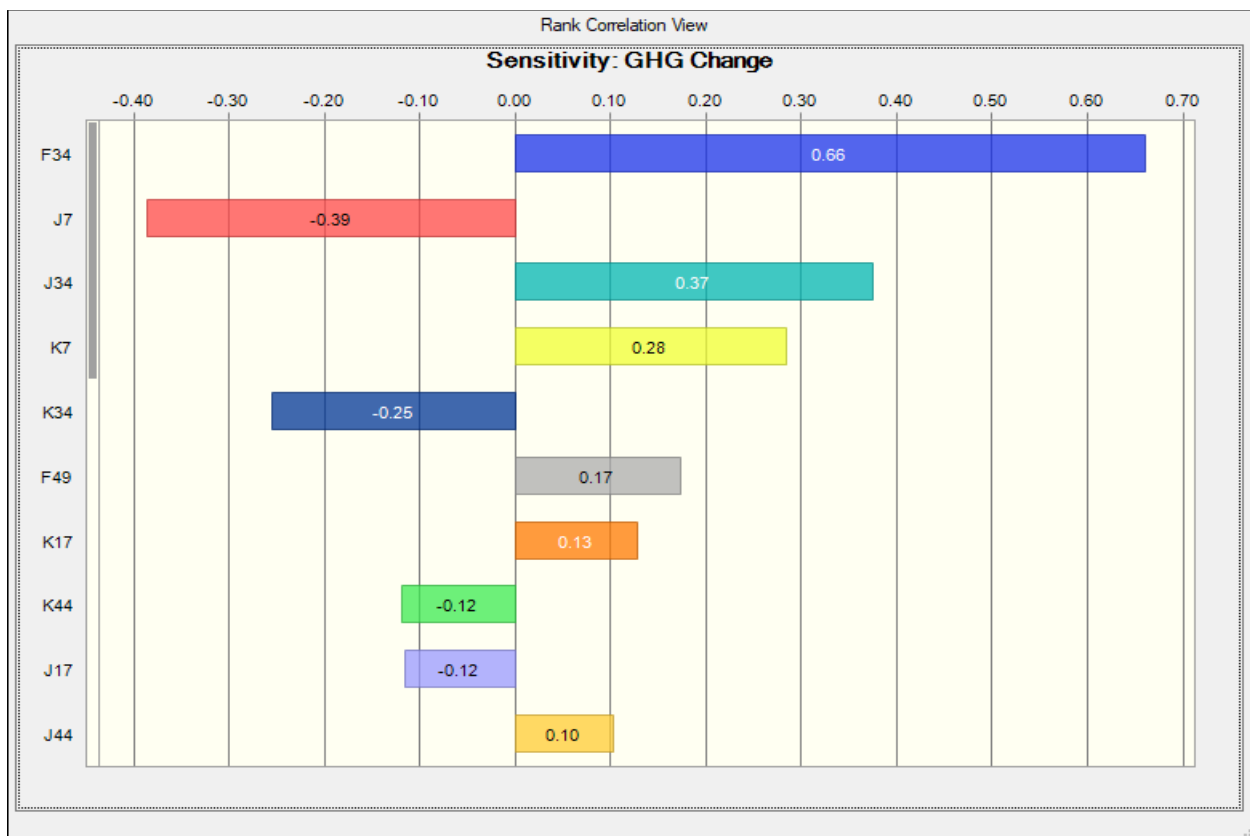
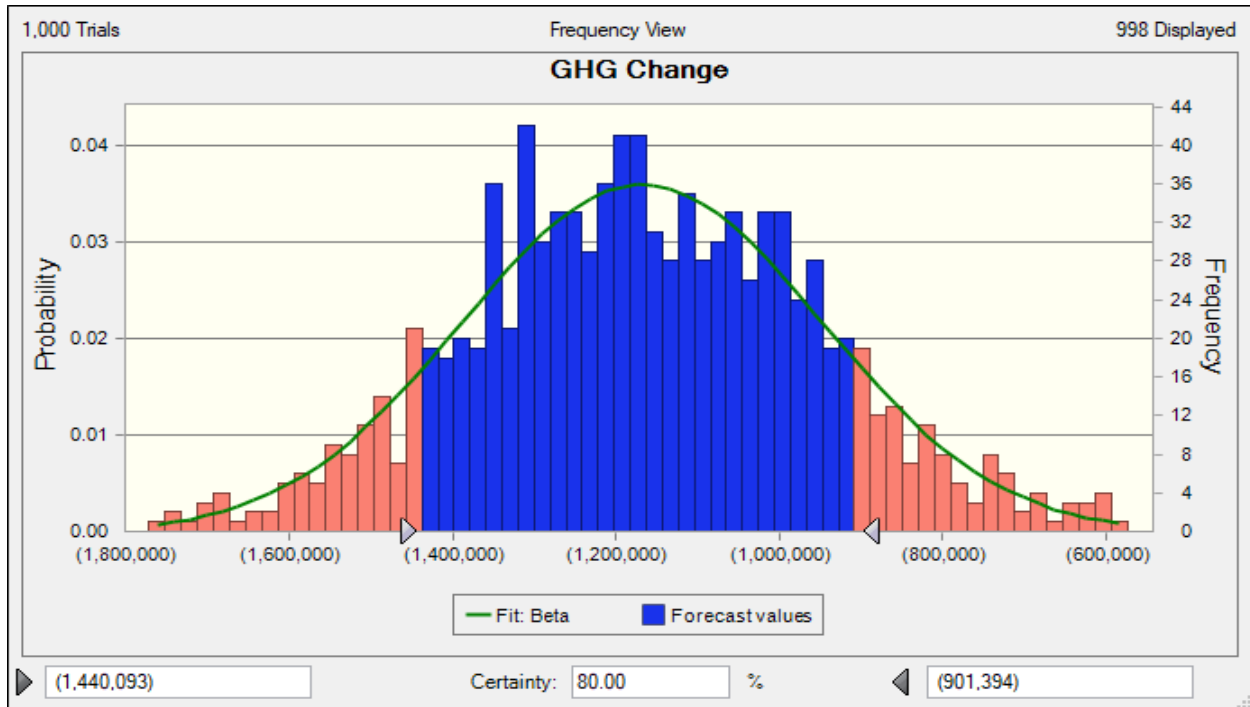
Variable	Definition	Source of Data
ASH(T)	Ash content of feed for cattle type T in the reporting period (see Table 5.3) (%)	Taken from Appendix 5-B of Powers et al. 2014. Additional feed values are available in the Powers Appendix.
B(T0)	Maximum methane-producing capacity of manure for cattle type T (see Table 5.2, m3 CH4 kg VS-1)	From Table 10.16 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use
BW(T)	Average live body weight (BW) of the animals in the population of cattle type T in kg	Table 10A.1 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
DE(T)	Diet digestibility of the feed for cattle type T in the reporting period as a fraction of gross energy (see Table 5.3) (dimensionless)	Taken from Appendix 5-B of Powers et al. 2014. Additional feed values are available in the Powers Appendix.
EF(dirS)	Emission factor for direct N2O emissions from non-anaerobic manure management system S (dimensionless) in Table 5.5.	Adapted from Table 10.21 in Chapter 10 of IPCC (2019).
EF(elec)	Emissions factor for electricity in the state in which the dairy is located (lbs CO2/MWh)	State-specific emissions factors are available via U.S. Energy Information Administration (2022). US Electricity Profile 2021. https://www.eia.gov/electricity/state/
EF(fuel,f)	Fuel-specific emission factor (kg/mmBtu or kg/gal)	Values calculated from data in U.S. Environmental Protection Agency. (2014). Emissions Factors for Greenhouse Gas Inventories. https://www.epa.gov/sites/default/files/2015-07/documents/emission-factors_2014.pdf using GWPCH4 = 29.8 and GWPN2O = 273 (Forster et al. 2021).
EF(l)	Emission factor for N2O emissions from nitrogen leaching and runoff.	Table 11.3 in Chapter 11 of IPCC (2019).
EF(V)	Emission factor for N2O emissions from nitrogen volatilization (kg N leaching/runoff)	Table 11.3 in Chapter 11 of IPCC (2019).
GWP(CH4)	Global warming potential of methane (kgCO2e kgCH4-1)	100-year time horizon; Chapter 7 of IPCC Sixth Assessment Report (Forster et al. 2021)
GWP(N2O)	Global warming potential of nitrous oxide (kg CO2e kg N2O-1)	100-year time horizon; Chapter 7 of IPCC Sixth Assessment Report (Forster et al. 2021)
MCF(S)	Methane conversion factor for non-anaerobic manure storage/treatment system S during the reporting period, based on T2, in Table 5.4 (%)	Values from CDFA (2023) AMMP Draft Benefits Calculator Tool.
MW(T)	Mature live body weight of an adult animal of cattle type T in moderate body condition in kg, either from producer records or using defaults	Based on 2018-2020 USDA NASS mean cow slaughter weight, assuming dressing % of 50%
N(Tex)	Average N excretion per animal of cattle type T (kg head-1)	Equation 10.31A in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

N(Tin)	Average daily nitrogen intake per animal of cattle type T in the reporting period (kg head ⁻¹ day ⁻¹)	35 Equation 10.32 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
N(Tret)	Fraction of daily N intake that is retained by cattle type T (dimensionless).	Equation 10.33 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
NE(G)	Net energy for growth (MJ day ⁻¹)	Equation 3-8 in NRC (1996)
PL(TS)	Percent of managed manure nitrogen losses for cattle type T due to runoff and leaching during solid and liquid storage of manure in non-anaerobic manure management system S (%)	From Table 10.22 in Chapter 10 of IPCC (2019).
PV(TS)	Percent of managed manure nitrogen for livestock type T that volatilizes as NH ₃ and NO _x from non-anaerobic manure management system S	From Table 10.22 in Chapter 10 of IPCC (2019).
RCH ₄	Density of methane	From Equation 10.23 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use
SCF	System calibration factor	From California Air Resources Board (2023) and US EPA (2016)
UE(T)	Urinary energy of cattle type T in the reporting period expressed as a fraction of gross energy (dimensionless).	From Equation 10.24 of Chapter 10: Emissions From Livestock and Manure Management. In 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4: Agriculture, Forestry and Other Land Use
WG(T)	Average weight gain of the animals in the population of cattle type T (kg day ⁻¹)	Table 10A.1 in 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

Appendix B. AMMP Uncertainty Deduction Model

Quantitative uncertainty in both the model and variables has been assessed with a Monte Carlo simulation. This choice in technique was based on both Gold Standard and IPCC methodologies for assessing uncertainty and will be added in the form of a new addition to the final GHG reduction equation in the form of an uncertainty deduction. Please see Section 5.1.3 for additional information.

Herd Description By Cattle Type	
Lactating (freestall)	1,000
Lactating (open lot)	-
Dry dairy cows	300
All other types	300
Emission Reduction (Herd - Kg CO ₂ e)	
Expected Reduction (ET ₀)	1,168,089
Uncertainty Percentile Boundaries:	
10th Percentile	(1,440,093)
90th Percentile	(901,394)
Uncertainty Deduction From Calculated Estimate	
Total Kg CO ₂ e Uncertainty Deduction (UNC)	0.56
90 % Confidence Uncertainty Deduction (UD)	656,583
GHG Emission Credits After Deduction	
Kg CO ₂ e	511,506
MT CO ₂ e	512

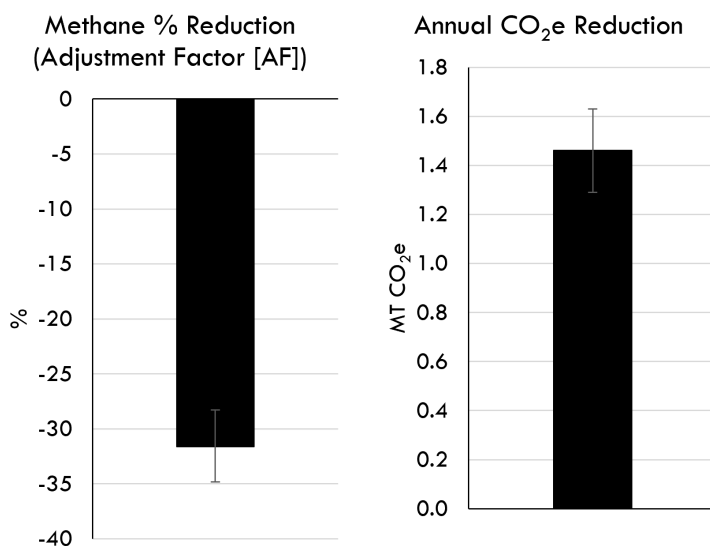


Appendix C. Bovaer Uncertainty Assessment

A quantitative uncertainty assessment for both the adjustment factor and across farm-level parameters was conducted. Table B reports uncertainty sources of all parameters used for Bovaer-related emissions quantifications. A description of the parameters with unknown certainty and their importance was assessed during validation. In accordance with GHG Protocol Scope 3 Quantitative Inventory Uncertainty guidance, a pedigree matrix approach was used to quantify uncertainty where a parameter uncertainty was unknown.

A Monte Carlo simulation was conducted to propagate parameter uncertainty to determine the overall uncertainty on the adjustment factor (AF) due to implementation of Bovaer and the parameters listed. This includes the SEM reported as ± 1.6 for the intercept, ± 0.074 for Dose, ± 0.252 for NDF parameters from Kebreab et al. (2023) and the values from the pedigree matrix described above. The Monte Carlo simulation for the adjustment factor yielded an expected methane percent change of -31.60%, accompanied by a 90% confidence interval ranging from -34.90% (upper bound) to -28.39% (lower bound) and an uncertainty of 10.3% (calculated as (upper bound – lower bound) / (2 x expected)). The Monte Carlo simulation for propagating uncertainty across all farm-level variables (and including the adjustment factor) equates to a mean CO₂e change (project minus baseline) of - 1.46 MT CO₂e with a 90% confidence interval ranging from -1.29 (lower bound) to -1.63 (upper bound) MT CO₂e annually and an uncertainty of 11.6%. Both are represented in Figure A below.

Figure A Impact assessment results of Bovaer implementation on methane reduction



Appendix D. Athian Dairy Total Digestible Nutrients Quantification Methodology

The methodology below was developed in conjunction with nutrition resources to determine the most streamlined methods with minimal data requirements from producers. There are two different calculation paths: one for lactating cattle and one for dry cows and heifers.

The Athian Dairy Total Digestible Nutrients Tool Methodology has been reviewed by the Athian Scientific Advisory Board and approved for use in this validated protocol.

D.1 Methodology for Lactating Cows

The lactating cattle methodology is based on the energy needs for maintenance and production of the lactating animals.

D.1.1 Total Digestible Nutrients²⁶

$$TDN = (DE_{intake}/0.0441)/100 \quad (\text{Equation 39})$$

Where:

TDN	=	Total digestible nutrients (%)
DE_{intake}	=	Dietary energy intake (Mcal/kg DMI), calculated per Equation 40.

D.1.2 Dietary Energy Intake

$$DE_{intake} = ME_{intake}/0.82 \quad (\text{Equation 40})$$

Where:

ME_{intake}	=	Metabolizable energy intake (Mcal/kg DMI), calculated per Equation 41.
0.82	=	Conversion factor from ME to DE ²⁷

D.1.3 Metabolizable Energy Intake

$$ME_{intake} = ME_{req}/DMI_T \quad (\text{Equation 41})$$

Where:

ME_{intake}	=	Metabolizable energy intake (Mcal/kg DMI), calculated per Equation 41.
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²⁶ NASEM 2001, Equation 2-1

²⁷ NRC 1996 via NASEM 2001, page 17

ME_{req}	=	Total required metabolizable energy (Mcal/day), calculated per Equation 42.
DMI_T	=	Dry matter intake of cattle type T (kg/hd/day)

D.1.4 Total Metabolizable Energy Required

$$ME_{req} = ME_{maint} + RE + ME_{lact} + ME_{growth} \quad (\text{Equation 42})$$

Where:

ME_{req}	=	Total required metabolizable energy (Mcal/day), calculated per Equation 42.
ME_{maint}	=	Metabolizable energy for maintenance (Mcal/day), calculated per Equation 44.
RE	=	Retained energy (Mcal/day), calculated per Equation 52. <i>Note: for lactating animals, this value is set to 0.</i>
ME_{lact}	=	Metabolizable energy for lactation (Mcal/day), calculated per Equation 43.
ME_{growth}	=	Metabolizable energy for growth (Mcal/day), calculated per Equation 53. <i>Note: for lactating animals this value is set to 0.</i>

D.1.5 Metabolizable Energy for Lactation²⁸

$$ME_{lact} = NE_{milk}/0.66 \quad (\text{Equation 43})$$

Where:

ME_{lact}	=	Metabolizable energy for lactation (Mcal/day), calculated per Equation 43.
NE_{milk}	=	Net energy requirements for milk production (total Mcals/hd/day), calculated per Equation 45.

D.1.6 Metabolizable Energy for Maintenance²⁹

$$ME_{maint} = BW^{0.75} \times 0.1 \quad (\text{Equation 44})$$

Where:

ME_{maint}	=	Metabolizable energy for maintenance (Mcal/day), calculated per Equation 44.
BW	=	Body weight (kg). See Table 1 for average live body weights by cohort.

²⁸ NASEM 2001, page 79

²⁹ NASEM 2021, Equation 3-13

Table D.1 Average Live Body Weights by Cohort

Cohort	Body Weight (kg)	Source
Lactating cows	650 kg	Derived from IPCC guidance from 2019
Dry cows	650 kg	Derived from IPCC guidance from 2019
Heifers	400 kg	Derived from IPCC guidance from 2019

D.1.7 Net Energy for Milk Production³⁰

$$NE_{milk} = (Fat \times 9.29) + (Protein \times 5.5) + (Lactose \times 3.95) \quad (Equation\ 45)$$

Where:

NE_{milk}	=	Net energy requirements for milk production (total Mcals/hd/day), calculated per Equation 45.
Fat	=	Milk fat content (kg fat/kg milk)
$Protein$	=	Milk protein crude content (kg crude protein/kg milk)
$Lactose$	=	Milk lactose content (kg lactose/kg milk), calculated per Equation 46.

D.1.8 Milk Lactose Content

$$Lactose = M \times 0.0485 \quad (Equation\ 46)$$

Where:

$Lactose$	=	Milk lactose content (kg lactose/kg milk), calculated per Equation 46.
M	=	Reported milk production (kg/hd/day). Note: reported milk production will most commonly be reported in lbs/hd/day and must be converted into kg.
0.0485	=	Estimated lactose content of milk (%) ³¹

D.2 Methodology for Dry Cows and Heifers

D.2.1 Total Digestible Nutrients³²

$$TDN = (DE_{intake}/0.0441)/100 \quad (Equation\ 47)$$

Where:

³⁰ NASEM 2021, Equation 3-14a

³¹ NASEM 2021, page 102

³² NASEM 2001, Equation 2-1

TDN	=	Total digestible nutrients (%)
DE_{intake}	=	Dietary energy intake (Mcal/kg DMI), calculated per Equation 48.

D.2.2 Dietary Energy Intake

$$DE_{intake} = ME_{intake} / 0.82 \quad (\text{Equation 48})$$

Where:

ME_{intake}	=	Metabolizable energy intake (Mcal/kg DMI), calculated per Equation 49.
0.82	=	Conversion factor from ME to DE ³³

D.2.3 Metabolizable Energy Intake

$$ME_{intake} = ME_{req} / DMI_T \quad (\text{Equation 49})$$

Where:

ME_{intake}	=	Metabolizable energy intake (Mcal/kg DMI), calculated per Equation 49.
ME_{req}	=	Total required metabolizable energy (Mcal/day), calculated per Equation 50.
DMI_T	=	Dry matter intake of cattle type T (kg/hd/day)

D.2.4 Total Metabolizable Energy Required

$$ME_{req} = ME_{maint} + RE + ME_{lact} + ME_{growth} \quad (\text{Equation 50})$$

Where:

ME_{req}	=	Total required metabolizable energy (Mcal/day), calculated per Equation 50.
ME_{maint}	=	Metabolizable energy for maintenance (Mcal/day), calculated per Equation 51.
RE	=	Retained energy (Mcal/day), calculated per Equation 52. <i>Note: for lactating animals, this value is set to 0.</i>
ME_{lact}	=	Metabolizable energy for lactation (Mcal/day), calculated per Equation 2.5.
ME_{growth}	=	Metabolizable energy for growth (Mcal/day), calculated per Equation 2.15. <i>Note: for lactating animals this value is set to 0.</i>

³³ NRC 1996 via NASEM 2001, page 17

D.2.5 Metabolizable Energy for Maintenance³⁴

$$ME_{maint} = BW^{0.75} \times 0.1 \quad (\text{Equation 51})$$

Where:

ME_{maint}	=	Metabolizable energy for maintenance (Mcals/day), calculated per Equation 51.
BW	=	Body weight (kg). See Table 1 for average live body weights by cohort.

D.2.6 Retained Energy³⁵

$$RE = 0.0635 \times BW^{0.75} \times ADG^{1.097} \quad (\text{Equation 52})$$

Where:

RE	=	Retained energy (Mcals/day), calculated per Equation 52. <i>Note: for lactating animals, this value is set to 0.</i>
BW	=	Body weight (kg). See Table C.1 for average live body weights by cohort.
ADG	=	Expected average daily gain (kg/hd/day). Default for dry cows is 0.17 ³⁶ and default for heifers is 0.55 ³⁷ .

D.2.7 Metabolizable Energy for Growth³⁸

$$ME_{growth} = RE/0.4 \quad (\text{Equation 53})$$

Where:

ME_{growth}	=	Metabolizable energy for growth (Mcals/day), calculated per Equation 53. <i>Note: for lactating animals this value is set to 0.</i>
RE	=	Retained energy (Mcals/day), calculated per Equation 52. <i>Note: for lactating animals, this value is set to 0.</i>

³⁴ NASEM 2021, Equation 3-13

³⁵ NASEM 2001, page 320. This equation was chosen over Equation 10-5 in NASEM 2021 to streamline data collection processes for producers.

³⁶ NASEM 2021, page 223, average of range

³⁷ NASEM 2021, page 516, average of range

³⁸ NASEM 2021, Equation 11-7